

Summary Tables of MAXCUT Data

1 Tables for depth $P = 1$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	10 [50]	-	9 [100]	7 [100]
Fixed Angle†	10 [50]	-	9 [100]	7 [100]
Fourier*	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angle*	10 [50]	-	9 [100]	7 [100]
Fixed Angle†	10 [50]	-	9 [100]	7 [100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	10 [50]	11 [106–144]	10 [100]	9 [40–100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
PP				
Fixed Angle*	10 [50]	-	9 [100]	7 [100]
Fixed Angle†	10 [50]	-	9 [100]	7 [100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	10 [50]	11 [106–144]	10 [100]	9 [40–100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
SV				
Fixed Angle*	64 [10–20]	-	19 [20]	273 [10–20]
Fixed Angle†	64 [10–20]	-	19 [20]	289 [10–20]
Fourier*	92 [10–20]	20 [12–21]	784 [10–20]	570 [10–20]
Interp.*	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	90 [10–20]	20 [12–21]	505 [10–22]	368 [10–20]
TQA	74 [10–20]	10 [21]	611 [10–20]	280 [10–20]
TQA*	72 [10–20]	10 [21]	603 [10–20]	66 [10–20]

Table 1: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	76.0 $\pm$ 1.2	-	72.4 $\pm$ 2.3	71.3 $\pm$ 1.3
Fixed Angle [†]	75.4 $\pm$ 1.3	-	72.2 $\pm$ 2.3	70.5 $\pm$ 1.6
Fourier [*]	81.5 $\pm$ 1.2	75.6 $\pm$ 1.2	73.1 $\pm$ 2.1	77.3 $\pm$ 2.1
Interp. [*]	81.5 $\pm$ 1.2	75.6 $\pm$ 1.2	73.1 $\pm$ 2.1	77.3 $\pm$ 2.1
Linear Ramp	75.6 $\pm$ 1.2	74.6 $\pm$ 0.9	71.8 $\pm$ 2.7	71.1 $\pm$ 1.3
Linear Ramp [*]	75.8 $\pm$ 1.3	74.6 $\pm$ 0.9	71.9 $\pm$ 2.7	71.2 $\pm$ 1.3
TQA	75.5 $\pm$ 1.3	74.5 $\pm$ 1.0	68.9 $\pm$ 4.3	71.0 $\pm$ 1.4
TQA [*]	75.6 $\pm$ 1.3	74.7 $\pm$ 1.0	71.8 $\pm$ 2.7	71.2 $\pm$ 1.4
MPS (Quimb)				
Fixed Angle [*]	75.6 $\pm$ 1.2	-	72.3 $\pm$ 2.2	69.3 $\pm$ 2.6
Fixed Angle [†]	74.9 $\pm$ 1.1	-	72.3 $\pm$ 2.2	68.1 $\pm$ 2.9
Fourier [*]	81.5 $\pm$ 1.2	75.4 $\pm$ 1.2	72.7 $\pm$ 1.7	77.1 $\pm$ 1.8
Interp. [*]	81.5 $\pm$ 1.2	75.4 $\pm$ 1.2	72.7 $\pm$ 1.7	77.1 $\pm$ 1.8
Linear Ramp	75.8 $\pm$ 1.1	68.2 $\pm$ 1.1	71.9 $\pm$ 2.6	69.2 $\pm$ 2.6
Linear Ramp [*]	75.8 $\pm$ 1.1	67.4 $\pm$ 1.3	71.9 $\pm$ 2.6	69.3 $\pm$ 2.5
Recursive TS	80.1 $\pm$ 1.1	74.6 $\pm$ 0.8	71.3 $\pm$ 2.0	77.2 $\pm$ 2.1
TQA	75.8 $\pm$ 1.2	67.3 $\pm$ 1.2	68.6 $\pm$ 4.5	68.9 $\pm$ 3.0
TQA [*]	75.8 $\pm$ 1.1	67.4 $\pm$ 1.3	71.9 $\pm$ 2.6	69.4 $\pm$ 2.4
PP				
Fixed Angle [*]	81.9 $\pm$ 1.1	-	73.0 $\pm$ 1.4	78.8 $\pm$ 2.2
Fixed Angle [†]	81.6 $\pm$ 1.0	-	73.0 $\pm$ 1.4	78.7 $\pm$ 2.2
Fourier [*]	81.5 $\pm$ 1.2	75.4 $\pm$ 1.2	72.7 $\pm$ 1.7	77.1 $\pm$ 1.8
Interp. [*]	81.5 $\pm$ 1.2	75.4 $\pm$ 1.2	72.7 $\pm$ 1.7	77.1 $\pm$ 1.8
Linear Ramp	81.9 $\pm$ 1.1	75.3 $\pm$ 1.0	72.6 $\pm$ 1.6	78.4 $\pm$ 2.3
Linear Ramp [*]	81.9 $\pm$ 1.1	75.3 $\pm$ 1.0	72.6 $\pm$ 1.6	78.4 $\pm$ 2.3
Recursive TS	80.1 $\pm$ 1.1	74.6 $\pm$ 0.8	71.3 $\pm$ 2.0	77.2 $\pm$ 2.1
TQA	77.8 $\pm$ 0.9	75.1 $\pm$ 0.9	68.9 $\pm$ 4.2	76.9 $\pm$ 1.0
TQA [*]	81.9 $\pm$ 1.1	75.4 $\pm$ 0.9	72.9 $\pm$ 1.4	78.8 $\pm$ 2.2
SV				
Fixed Angle [*]	79.0 $\pm$ 2.8	-	73.4 $\pm$ 2.6	81.7 $\pm$ 4.3
Fixed Angle [†]	78.4 $\pm$ 2.5	-	73.4 $\pm$ 2.6	80.7 $\pm$ 3.1
Fourier [*]	78.8 $\pm$ 2.8	78.3 $\pm$ 2.6	74.7 $\pm$ 4.0	82.0 $\pm$ 4.1
Interp. [*]	78.8 $\pm$ 2.8	78.3 $\pm$ 2.6	74.6 $\pm$ 3.9	81.9 $\pm$ 4.2
Linear Ramp	78.4 $\pm$ 1.9	77.4 $\pm$ 1.7	74.2 $\pm$ 1.9	79.9 $\pm$ 2.8
Recursive TS	78.8 $\pm$ 2.8	78.3 $\pm$ 2.6	74.9 $\pm$ 4.8	82.0 $\pm$ 4.3
TQA	77.6 $\pm$ 2.5	75.9 $\pm$ 2.7	71.0 $\pm$ 5.0	79.0 $\pm$ 2.9
TQA [*]	78.7 $\pm$ 2.9	77.4 $\pm$ 1.7	74.7 $\pm$ 3.6	78.3 $\pm$ 2.2

Table 2: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	76.0 ± 1.2	-	72.4 ± 2.3	71.3 ± 1.3	10 [50]	-	9 [100]	7 [100]
Fixed Angle†	75.4 ± 1.3	-	72.2 ± 2.3	70.5 ± 1.6	10 [50]	-	9 [100]	7 [100]
Fourier*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Interp.*	81.5 ± 1.2	75.6 ± 1.2	73.1 ± 2.1	77.3 ± 2.1	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Linear Ramp	75.6 ± 1.2	74.6 ± 0.9	71.8 ± 2.7	71.1 ± 1.3	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	75.8 ± 1.3	74.6 ± 0.9	71.9 ± 2.7	71.2 ± 1.3	10 [50]	10 [144]	10 [100]	7 [100]
TQA	75.5 ± 1.3	74.5 ± 1.0	68.9 ± 4.3	71.0 ± 1.4	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	75.6 ± 1.3	74.7 ± 1.0	71.8 ± 2.7	71.2 ± 1.4	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)								
Fixed Angle*	75.6 ± 1.2	-	72.3 ± 2.2	69.3 ± 2.6	10 [50]	-	9 [100]	7 [100]
Fixed Angle†	74.9 ± 1.1	-	72.3 ± 2.2	68.1 ± 2.9	10 [50]	-	9 [100]	7 [100]
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp	75.8 ± 1.1	68.2 ± 1.1	71.9 ± 2.6	69.2 ± 2.6	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.3 ± 2.5	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	80.1 ± 1.1	74.6 ± 0.8	71.3 ± 2.0	77.2 ± 2.1	10 [50]	11 [106–144]	10 [100]	9 [40–100]
TQA	75.8 ± 1.2	67.3 ± 1.2	68.6 ± 4.5	68.9 ± 3.0	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	75.8 ± 1.1	67.4 ± 1.3	71.9 ± 2.6	69.4 ± 2.4	10 [50]	10 [144]	10 [100]	7 [100]
PP								
Fixed Angle*	81.9 ± 1.1	-	73.0 ± 1.4	78.8 ± 2.2	10 [50]	-	9 [100]	7 [100]
Fixed Angle†	81.6 ± 1.0	-	73.0 ± 1.4	78.7 ± 2.2	10 [50]	-	9 [100]	7 [100]
Fourier*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Interp.*	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	77.1 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	58 [30–100]
Linear Ramp	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp*	81.9 ± 1.1	75.3 ± 1.0	72.6 ± 1.6	78.4 ± 2.3	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	80.1 ± 1.1	74.6 ± 0.8	71.3 ± 2.0	77.2 ± 2.1	10 [50]	11 [106–144]	10 [100]	9 [40–100]
TQA	77.8 ± 0.9	75.1 ± 0.9	68.9 ± 4.2	76.9 ± 1.0	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	81.9 ± 1.1	75.4 ± 0.9	72.9 ± 1.4	78.8 ± 2.2	10 [50]	10 [144]	10 [100]	7 [100]
SV								
Fixed Angle*	79.0 ± 2.8	-	73.4 ± 2.6	81.7 ± 4.3	64 [10–20]	-	19 [20]	273 [10–20]
Fixed Angle†	78.4 ± 2.5	-	73.4 ± 2.6	80.7 ± 3.1	64 [10–20]	-	19 [20]	289 [10–20]
Fourier*	78.8 ± 2.8	78.3 ± 2.6	74.7 ± 4.0	82.0 ± 4.1	92 [10–20]	20 [12–21]	784 [10–20]	570 [10–20]
Interp.*	78.8 ± 2.8	78.3 ± 2.6	74.6 ± 3.9	81.9 ± 4.2	92 [10–20]	20 [12–21]	828 [10–22]	382 [10–20]
Linear Ramp	78.4 ± 1.9	77.4 ± 1.7	74.2 ± 1.9	79.9 ± 2.8	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	78.8 ± 2.8	78.3 ± 2.6	74.9 ± 4.8	82.0 ± 4.3	90 [10–20]	20 [12–21]	505 [10–22]	368 [10–20]
TQA	77.6 ± 2.5	75.9 ± 2.7	71.0 ± 5.0	79.0 ± 2.9	74 [10–20]	10 [21]	611 [10–20]	280 [10–20]
TQA*	78.7 ± 2.9	77.4 ± 1.7	74.7 ± 3.6	78.3 ± 2.2	72 [10–20]	10 [21]	603 [10–20]	66 [10–20]

Table 3: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

2 Tables for depth $P = 2$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	10 [50]	10 [39]	27 [100]	17 [40-100]
Fixed Angle†	10 [50]	10 [39]	27 [100]	17 [40-100]
Fourier*	20 [40-50]	20 [39-144]	55 [40-100]	17 [40-100]
Interp.*	20 [40-50]	20 [39-144]	55 [40-100]	17 [40-100]
Linear Ramp	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	20 [40-50]	20 [39-144]	56 [40-100]	17 [40-100]
TQA*	20 [40-50]	20 [39-144]	56 [40-100]	17 [40-100]
MPS (Quimb)				
Fixed Angle*	10 [50]	10 [144]	90 [100]	47 [30-100]
Fixed Angle†	10 [50]	10 [144]	90 [100]	47 [30-100]
Fourier*	20 [40-50]	31 [39-144]	130 [40-100]	58 [30-100]
Interp.*	20 [40-50]	31 [39-144]	130 [40-100]	58 [30-100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-
TQA	20 [40-50]	20 [39-144]	130 [40-100]	57 [30-100]
TQA*	20 [40-50]	20 [39-144]	130 [40-100]	57 [30-100]
PP				
Fixed Angle*	10 [50]	30 [39-144]	82 [100]	47 [30-100]
Fixed Angle†	10 [50]	30 [39-144]	82 [100]	47 [30-100]
Fourier*	20 [40-50]	31 [39-144]	130 [40-100]	58 [30-100]
Interp.*	20 [40-50]	31 [39-144]	130 [40-100]	58 [30-100]
Linear Ramp	10 [50]	20 [105-144]	29 [40-100]	8 [100]
Linear Ramp*	10 [50]	20 [105-144]	20 [40-100]	8 [100]
Recursive TS	-	-	-	-
TQA	20 [40-50]	20 [39-144]	130 [40-100]	57 [30-100]
TQA*	20 [40-50]	20 [39-144]	130 [40-100]	57 [30-100]
SV				
Fixed Angle*	81 [10-20]	-	311 [10-22]	360 [10-20]
Fixed Angle†	81 [10-20]	-	311 [10-22]	372 [10-20]
Fourier*	92 [10-20]	20 [12-21]	784 [10-20]	567 [10-20]
Interp.*	92 [10-20]	20 [12-21]	828 [10-22]	372 [10-20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	90 [10-20]	20 [12-21]	425 [10-22]	353 [10-20]
TQA	92 [10-20]	20 [12-21]	822 [10-22]	361 [10-20]
TQA*	92 [10-20]	20 [12-21]	821 [10-22]	142 [10-20]

Table 4: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	77.0 ± 1.9	82.2 ± 1.3	75.3 ± 3.9	75.6 ± 4.9
Fixed Angle†	75.6 ± 1.3	80.9 ± 1.5	70.6 ± 6.4	74.7 ± 5.1
Fourier*	71.5 ± 2.3	75.6 ± 3.8	69.3 ± 4.2	73.1 ± 3.5
Interp.*	77.0 ± 1.6	80.4 ± 2.0	73.1 ± 5.2	70.9 ± 2.2
Linear Ramp	76.2 ± 1.3	77.7 ± 1.5	73.0 ± 5.8	70.1 ± 2.2
Linear Ramp*	77.9 ± 2.1	77.9 ± 1.2	78.4 ± 4.3	70.1 ± 2.2
TQA	69.0 ± 1.7	72.7 ± 1.4	67.1 ± 3.9	72.1 ± 2.8
TQA*	76.0 ± 1.2	78.8 ± 2.9	72.2 ± 6.2	74.4 ± 4.6
MPS (Quimb)				
Fixed Angle*	80.5 ± 1.9	53.9 ± 0.6	79.9 ± 2.4	76.4 ± 4.9
Fixed Angle†	77.2 ± 0.9	52.5 ± 0.6	79.6 ± 2.6	75.3 ± 5.5
Fourier*	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	74.6 ± 4.7
Interp.*	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	74.9 ± 4.4
Linear Ramp	78.5 ± 3.1	69.5 ± 1.3	79.0 ± 3.3	70.2 ± 4.0
Linear Ramp*	79.3 ± 2.5	68.7 ± 1.4	79.6 ± 3.0	70.4 ± 5.3
Recursive TS	-	-	-	-
TQA	69.3 ± 1.1	63.5 ± 9.8	70.0 ± 3.9	72.2 ± 3.0
TQA*	77.4 ± 1.3	68.6 ± 14.9	79.2 ± 3.0	76.5 ± 4.8
PP				
Fixed Angle*	87.8 ± 1.0	82.7 ± 1.2	79.2 ± 1.6	83.8 ± 1.5
Fixed Angle†	86.9 ± 1.0	81.2 ± 1.4	79.2 ± 1.6	83.7 ± 1.5
Fourier*	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.5 ± 2.7
Interp.*	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4
Linear Ramp	86.9 ± 1.0	82.4 ± 1.0	80.3 ± 1.7	83.9 ± 1.2
Linear Ramp*	87.8 ± 1.0	82.5 ± 1.0	80.3 ± 1.9	83.9 ± 1.1
Recursive TS	-	-	-	-
TQA	78.9 ± 1.3	72.8 ± 1.4	70.6 ± 3.0	77.1 ± 2.0
TQA*	87.4 ± 1.2	82.8 ± 1.4	79.6 ± 2.0	83.8 ± 1.4
SV				
Fixed Angle*	85.4 ± 2.2	-	82.5 ± 2.9	88.1 ± 3.3
Fixed Angle†	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0
Fourier*	76.4 ± 8.8	75.1 ± 4.5	76.2 ± 5.7	80.6 ± 8.9
Interp.*	85.3 ± 2.3	84.8 ± 1.7	82.5 ± 4.0	86.5 ± 4.6
Linear Ramp	84.6 ± 1.7	84.5 ± 1.2	81.7 ± 1.8	85.7 ± 2.4
Recursive TS	85.2 ± 2.4	84.3 ± 3.3	81.3 ± 2.8	86.1 ± 4.3
TQA	76.2 ± 3.1	72.9 ± 4.2	72.5 ± 4.2	75.7 ± 4.4
TQA*	85.5 ± 2.3	85.7 ± 2.1	82.8 ± 3.1	87.3 ± 3.3

Table 5: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	77.0 ± 1.9	82.2 ± 1.3	75.3 ± 3.9	75.6 ± 4.9	10 [50]	10 [39]	27 [100]	17 [40-100]
Fixed Angle†	75.6 ± 1.3	80.9 ± 1.5	70.6 ± 6.4	74.7 ± 5.1	10 [50]	10 [39]	27 [100]	17 [40-100]
Fourier*	71.5 ± 2.3	75.6 ± 3.8	69.3 ± 4.2	73.1 ± 3.5	20 [40-50]	20 [39-144]	55 [40-100]	17 [40-100]
Interp.*	77.0 ± 1.6	80.4 ± 2.0	73.1 ± 5.2	70.9 ± 2.2	20 [40-50]	20 [39-144]	55 [40-100]	17 [40-100]
Linear Ramp	76.2 ± 1.3	77.7 ± 1.5	73.0 ± 5.8	70.1 ± 2.2	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp*	77.9 ± 2.1	77.9 ± 1.2	78.4 ± 4.3	70.1 ± 2.2	10 [50]	10 [144]	10 [100]	7 [100]
TQA	69.0 ± 1.7	72.7 ± 1.4	67.1 ± 3.9	72.1 ± 2.8	20 [40-50]	20 [39-144]	56 [40-100]	17 [40-100]
TQA*	76.0 ± 1.2	78.8 ± 2.9	72.2 ± 6.2	74.4 ± 4.6	20 [40-50]	20 [39-144]	56 [40-100]	17 [40-100]
MPS (Quimb)								
Fixed Angle*	80.5 ± 1.9	53.9 ± 0.6	79.9 ± 2.4	76.4 ± 4.9	10 [50]	10 [144]	90 [100]	47 [30-100]
Fixed Angle†	77.2 ± 0.9	52.5 ± 0.6	79.6 ± 2.6	75.3 ± 5.5	10 [50]	10 [144]	90 [100]	47 [30-100]
Fourier*	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	74.6 ± 4.7	20 [40-50]	31 [39-144]	130 [40-100]	58 [30-100]
Interp.*	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	74.9 ± 4.4	20 [40-50]	31 [39-144]	130 [40-100]	58 [30-100]
Linear Ramp	78.5 ± 3.1	69.5 ± 1.3	79.0 ± 3.3	70.2 ± 4.0	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	79.3 ± 2.5	68.7 ± 1.4	79.6 ± 3.0	70.4 ± 5.3	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-	-	-	-	-
TQA	69.3 ± 1.1	63.5 ± 9.8	70.0 ± 3.9	72.2 ± 3.0	20 [40-50]	20 [39-144]	130 [40-100]	57 [30-100]
TQA*	77.4 ± 1.3	68.6 ± 14.9	79.2 ± 3.0	76.5 ± 4.8	20 [40-50]	20 [39-144]	130 [40-100]	57 [30-100]
PP								
Fixed Angle*	87.8 ± 1.0	82.7 ± 1.2	79.2 ± 1.6	83.8 ± 1.5	10 [50]	30 [39-144]	82 [100]	47 [30-100]
Fixed Angle†	86.9 ± 1.0	81.2 ± 1.4	79.2 ± 1.6	83.7 ± 1.5	10 [50]	30 [39-144]	82 [100]	47 [30-100]
Fourier*	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.5 ± 2.7	20 [40-50]	31 [39-144]	130 [40-100]	58 [30-100]
Interp.*	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4	20 [40-50]	31 [39-144]	130 [40-100]	58 [30-100]
Linear Ramp	86.9 ± 1.0	82.4 ± 1.0	80.3 ± 1.7	83.9 ± 1.2	10 [50]	20 [105-144]	29 [40-100]	8 [100]
Linear Ramp*	87.8 ± 1.0	82.5 ± 1.0	80.3 ± 1.9	83.9 ± 1.1	10 [50]	20 [105-144]	20 [40-100]	8 [100]
Recursive TS	-	-	-	-	-	-	-	-
TQA	78.9 ± 1.3	72.8 ± 1.4	70.6 ± 3.0	77.1 ± 2.0	20 [40-50]	20 [39-144]	130 [40-100]	57 [30-100]
TQA*	87.4 ± 1.2	82.8 ± 1.4	79.6 ± 2.0	83.8 ± 1.4	20 [40-50]	20 [39-144]	130 [40-100]	57 [30-100]
SV								
Fixed Angle*	85.4 ± 2.2	-	82.5 ± 2.9	88.1 ± 3.3	81 [10-20]	-	311 [10-22]	360 [10-22]
Fixed Angle†	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0	81 [10-20]	-	311 [10-22]	372 [10-22]
Fourier*	76.4 ± 8.8	75.1 ± 4.5	76.2 ± 5.7	80.6 ± 8.9	92 [10-20]	20 [12-21]	784 [10-20]	567 [10-22]
Interp.*	85.3 ± 2.3	84.8 ± 1.7	82.5 ± 4.0	86.5 ± 4.6	92 [10-20]	20 [12-21]	828 [10-22]	372 [10-22]
Linear Ramp	84.6 ± 1.7	84.5 ± 1.2	81.7 ± 1.8	85.7 ± 2.4	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	85.2 ± 2.4	84.3 ± 3.3	81.3 ± 2.8	86.1 ± 4.3	90 [10-20]	20 [12-21]	425 [10-22]	353 [10-22]
TQA	76.2 ± 3.1	72.9 ± 4.2	72.5 ± 4.2	75.7 ± 4.4	92 [10-20]	20 [12-21]	822 [10-22]	361 [10-22]
TQA*	85.5 ± 2.3	85.7 ± 2.1	82.8 ± 3.1	87.3 ± 3.3	92 [10-20]	20 [12-21]	821 [10-22]	142 [10-22]

Table 6: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

3 Tables for depth $P = 3$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	7 [50]	-	8 [100]	7 [100]
Fixed Angle†	7 [50]	-	8 [100]	7 [100]
Fourier*	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angle*	7 [50]	10 [144]	8 [100]	37 [30–100]
Fixed Angle†	7 [50]	10 [144]	8 [100]	37 [30–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-
TQA	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA*	10 [50]	10 [144]	20 [40–100]	37 [30–100]
PP				
Fixed Angle*	7 [50]	10 [144]	8 [100]	37 [30–100]
Fixed Angle†	7 [50]	10 [144]	8 [100]	37 [30–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	-	-	-	-
TQA	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA*	10 [50]	10 [144]	20 [40–100]	37 [30–100]
SV				
Fixed Angle*	81 [10–20]	-	283 [10–20]	365 [10–20]
Fixed Angle†	81 [10–20]	-	283 [10–20]	372 [10–20]
Fourier*	91 [10–20]	20 [12–21]	784 [10–20]	567 [10–20]
Interp.*	91 [10–20]	20 [12–21]	827 [10–22]	372 [10–20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	90 [10–20]	20 [12–21]	431 [10–22]	357 [10–20]
TQA	92 [10–20]	20 [12–21]	821 [10–22]	357 [10–20]
TQA*	91 [10–20]	20 [12–21]	821 [10–22]	141 [10–20]

Table 7: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	78.7 ± 3.5	-	87.9 ± 1.3	72.6 ± 4.6
Fixed Angle [†]	75.2 ± 1.4	-	84.6 ± 3.9	68.6 ± 2.3
Fourier [*]	79.9 ± 4.0	77.9 ± 2.7	71.8 ± 7.4	74.5 ± 4.0
Interp. [*]	81.1 ± 3.7	83.4 ± 3.0	80.0 ± 7.0	72.3 ± 3.3
Linear Ramp	76.9 ± 2.0	77.4 ± 1.3	85.7 ± 1.6	70.2 ± 2.8
Linear Ramp [*]	78.6 ± 3.6	79.3 ± 1.2	87.9 ± 1.4	70.9 ± 3.2
TQA	69.4 ± 3.9	76.7 ± 2.0	78.5 ± 5.8	69.8 ± 2.5
TQA [*]	75.9 ± 1.2	78.6 ± 2.5	82.9 ± 4.7	70.9 ± 3.3
MPS (Quimb)				
Fixed Angle [*]	84.5 ± 2.2	56.5 ± 1.5	86.4 ± 1.7	81.9 ± 6.0
Fixed Angle [†]	79.0 ± 2.1	54.1 ± 0.8	85.7 ± 2.3	79.1 ± 7.8
Fourier [*]	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	77.0 ± 5.6
Interp. [*]	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	80.5 ± 6.9
Linear Ramp	85.5 ± 2.2	71.4 ± 1.2	84.1 ± 1.4	74.7 ± 4.8
Linear Ramp [*]	88.7 ± 2.5	70.7 ± 1.2	86.9 ± 1.4	76.3 ± 5.1
Recursive TS	-	-	-	-
TQA	75.3 ± 3.6	57.0 ± 0.9	80.5 ± 3.3	74.7 ± 4.7
TQA [*]	81.3 ± 5.0	57.5 ± 1.0	86.4 ± 3.7	79.5 ± 7.3
PP				
Fixed Angle [*]	90.8 ± 1.2	86.4 ± 1.0	83.8 ± 1.6	87.4 ± 1.4
Fixed Angle [†]	89.7 ± 1.2	83.7 ± 1.5	83.0 ± 1.2	87.4 ± 1.3
Fourier [*]	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.7 ± 2.1
Interp. [*]	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.2 ± 1.4
Linear Ramp	89.6 ± 1.0	85.4 ± 0.8	82.3 ± 1.6	85.8 ± 1.2
Linear Ramp [*]	91.2 ± 1.1	86.5 ± 0.9	84.2 ± 1.9	86.8 ± 0.9
Recursive TS	-	-	-	-
TQA	80.7 ± 0.8	79.1 ± 1.0	71.4 ± 4.6	81.0 ± 1.5
TQA [*]	89.4 ± 1.4	83.5 ± 0.8	80.0 ± 2.1	86.9 ± 1.6
SV				
Fixed Angle [*]	89.1 ± 1.8	-	87.7 ± 2.6	91.6 ± 2.8
Fixed Angle [†]	87.1 ± 1.7	-	87.2 ± 2.5	89.9 ± 2.2
Fourier [*]	83.7 ± 5.7	79.8 ± 5.3	78.7 ± 7.5	88.5 ± 4.1
Interp. [*]	88.8 ± 1.9	89.0 ± 1.6	87.4 ± 3.8	89.7 ± 4.1
Linear Ramp	87.7 ± 1.5	86.8 ± 1.8	84.9 ± 1.8	88.5 ± 2.3
Recursive TS	87.8 ± 2.5	87.3 ± 3.3	84.9 ± 2.4	88.5 ± 3.8
TQA	81.3 ± 3.8	79.5 ± 4.1	77.3 ± 5.1	80.1 ± 5.5
TQA [*]	87.7 ± 2.2	87.0 ± 2.1	85.1 ± 3.8	89.8 ± 2.6

Table 8: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	78.7 ± 3.5	-	87.9 ± 1.3	72.6 ± 4.6	7 [50]	-	8 [100]	7 [100]
Fixed Angle†	75.2 ± 1.4	-	84.6 ± 3.9	68.6 ± 2.3	7 [50]	-	8 [100]	7 [100]
Fourier*	79.9 ± 4.0	77.9 ± 2.7	71.8 ± 7.4	74.5 ± 4.0	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Interp.*	81.1 ± 3.7	83.4 ± 3.0	80.0 ± 7.0	72.3 ± 3.3	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Linear Ramp	76.9 ± 2.0	77.4 ± 1.3	85.7 ± 1.6	70.2 ± 2.8	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	78.6 ± 3.6	79.3 ± 1.2	87.9 ± 1.4	70.9 ± 3.2	10 [50]	10 [144]	10 [100]	7 [100]
TQA	69.4 ± 3.9	76.7 ± 2.0	78.5 ± 5.8	69.8 ± 2.5	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	75.9 ± 1.2	78.6 ± 2.5	82.9 ± 4.7	70.9 ± 3.3	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)								
Fixed Angle*	84.5 ± 2.2	56.5 ± 1.5	86.4 ± 1.7	81.9 ± 6.0	7 [50]	10 [144]	8 [100]	37 [30–100]
Fixed Angle†	79.0 ± 2.1	54.1 ± 0.8	85.7 ± 2.3	79.1 ± 7.8	7 [50]	10 [144]	8 [100]	37 [30–100]
Fourier*	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	77.0 ± 5.6	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Interp.*	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	80.5 ± 6.9	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Linear Ramp	85.5 ± 2.2	71.4 ± 1.2	84.1 ± 1.4	74.7 ± 4.8	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	88.7 ± 2.5	70.7 ± 1.2	86.9 ± 1.4	76.3 ± 5.1	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-	-	-	-	-
TQA	75.3 ± 3.6	57.0 ± 0.9	80.5 ± 3.3	74.7 ± 4.7	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA*	81.3 ± 5.0	57.5 ± 1.0	86.4 ± 3.7	79.5 ± 7.3	10 [50]	10 [144]	20 [40–100]	37 [30–100]
PP								
Fixed Angle*	90.8 ± 1.2	86.4 ± 1.0	83.8 ± 1.6	87.4 ± 1.4	7 [50]	10 [144]	8 [100]	37 [30–100]
Fixed Angle†	89.7 ± 1.2	83.7 ± 1.5	83.0 ± 1.2	87.4 ± 1.3	7 [50]	10 [144]	8 [100]	37 [30–100]
Fourier*	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.7 ± 2.1	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Interp.*	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.2 ± 1.4	20 [40–50]	31 [39–144]	130 [40–100]	48 [30–100]
Linear Ramp	89.6 ± 1.0	85.4 ± 0.8	82.3 ± 1.6	85.8 ± 1.2	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp*	91.2 ± 1.1	86.5 ± 0.9	84.2 ± 1.9	86.8 ± 0.9	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	-	-	-	-	-	-	-	-
TQA	80.7 ± 0.8	79.1 ± 1.0	71.4 ± 4.6	81.0 ± 1.5	10 [50]	10 [144]	20 [40–100]	37 [30–100]
TQA*	89.4 ± 1.4	83.5 ± 0.8	80.0 ± 2.1	86.9 ± 1.6	10 [50]	10 [144]	20 [40–100]	37 [30–100]
SV								
Fixed Angle*	89.1 ± 1.8	-	87.7 ± 2.6	91.6 ± 2.8	81 [10–20]	-	283 [10–20]	365 [10–20]
Fixed Angle†	87.1 ± 1.7	-	87.2 ± 2.5	89.9 ± 2.2	81 [10–20]	-	283 [10–20]	372 [10–20]
Fourier*	83.7 ± 5.7	79.8 ± 5.3	78.7 ± 7.5	88.5 ± 4.1	91 [10–20]	20 [12–21]	784 [10–20]	567 [10–20]
Interp.*	88.8 ± 1.9	89.0 ± 1.6	87.4 ± 3.8	89.7 ± 4.1	91 [10–20]	20 [12–21]	827 [10–22]	372 [10–20]
Linear Ramp	87.7 ± 1.5	86.8 ± 1.8	84.9 ± 1.8	88.5 ± 2.3	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	87.8 ± 2.5	87.3 ± 3.3	84.9 ± 2.4	88.5 ± 3.8	90 [10–20]	20 [12–21]	431 [10–22]	357 [10–20]
TQA	81.3 ± 3.8	79.5 ± 4.1	77.3 ± 5.1	80.1 ± 5.5	92 [10–20]	20 [12–21]	821 [10–22]	357 [10–20]
TQA*	87.7 ± 2.2	87.0 ± 2.1	85.1 ± 3.8	89.8 ± 2.6	91 [10–20]	20 [12–21]	821 [10–22]	141 [10–20]

Table 9: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

4 Tables for depth $P = 4$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	10 [39]	21 [100]	13 [40-100]
Fixed Angle†	-	10 [39]	21 [100]	13 [40-100]
Fourier*	20 [40-50]	20 [39-144]	55 [40-100]	17 [40-100]
Interp.*	20 [40-50]	20 [39-144]	55 [40-100]	17 [40-100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	20 [40-50]	20 [39-144]	56 [40-100]	17 [40-100]
TQA*	20 [40-50]	20 [39-144]	56 [40-100]	17 [40-100]
MPS (Quimb)				
Fixed Angle*	-	10 [144]	30 [100]	13 [40-100]
Fixed Angle†	-	10 [144]	30 [100]	13 [40-100]
Fourier*	20 [40-50]	31 [39-144]	130 [40-100]	28 [40-100]
Interp.*	20 [40-50]	31 [39-144]	130 [40-100]	28 [40-100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-
TQA	20 [40-50]	20 [39-144]	130 [40-100]	27 [40-100]
TQA*	20 [40-50]	20 [39-144]	130 [40-100]	27 [40-100]
PP				
Fixed Angle*	-	30 [39-144]	21 [100]	13 [40-100]
Fixed Angle†	-	30 [39-144]	21 [100]	13 [40-100]
Fourier*	20 [40-50]	31 [39-144]	130 [40-100]	28 [40-100]
Interp.*	20 [40-50]	31 [39-144]	130 [40-100]	28 [40-100]
Linear Ramp	10 [50]	20 [105-144]	20 [40-100]	8 [100]
Linear Ramp*	10 [50]	20 [105-144]	20 [40-100]	8 [100]
Recursive TS	-	-	-	-
TQA	20 [40-50]	20 [39-144]	130 [40-100]	27 [40-100]
TQA*	20 [40-50]	20 [39-144]	130 [40-100]	27 [40-100]
SV				
Fixed Angle*	61 [10-20]	-	106 [10-20]	155 [10-20]
Fixed Angle†	61 [10-20]	-	106 [10-20]	158 [10-20]
Fourier*	91 [10-20]	20 [12-21]	784 [10-20]	567 [10-20]
Interp.*	91 [10-20]	20 [12-21]	827 [10-22]	372 [10-20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	90 [10-20]	20 [12-21]	430 [10-22]	353 [10-20]
TQA	92 [10-20]	20 [12-21]	821 [10-22]	355 [10-20]
TQA*	91 [10-20]	20 [12-21]	821 [10-22]	142 [10-20]

Table 10: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	88.8 ± 1.3	82.8 ± 6.9	88.6 ± 8.8
Fixed Angle [†]	-	87.9 ± 1.5	77.1 ± 8.3	85.4 ± 11.0
Fourier*	75.6 ± 4.5	80.9 ± 2.7	74.4 ± 8.1	76.6 ± 5.5
Interp.*	87.0 ± 2.6	85.7 ± 3.4	86.1 ± 6.2	77.7 ± 6.7
Linear Ramp	80.1 ± 4.6	79.9 ± 1.7	88.6 ± 1.9	72.8 ± 5.2
Linear Ramp*	85.8 ± 4.1	81.4 ± 1.0	91.1 ± 1.2	75.0 ± 6.0
TQA	69.6 ± 1.8	79.8 ± 2.1	74.4 ± 9.2	75.4 ± 5.1
TQA*	77.8 ± 4.9	82.5 ± 2.7	78.7 ± 9.6	79.2 ± 6.3
MPS (Quimb)				
Fixed Angle*	-	52.8 ± 1.1	90.5 ± 1.4	89.5 ± 5.9
Fixed Angle [†]	-	51.0 ± 0.9	90.4 ± 1.5	84.8 ± 10.7
Fourier*	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	83.6 ± 9.6
Interp.*	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	87.3 ± 8.4
Linear Ramp	90.5 ± 1.5	69.3 ± 2.0	87.7 ± 1.5	75.7 ± 5.4
Linear Ramp*	92.4 ± 1.9	68.8 ± 2.0	90.2 ± 1.0	77.8 ± 7.4
Recursive TS	-	-	-	-
TQA	70.5 ± 3.4	68.1 ± 14.3	84.2 ± 2.4	77.4 ± 5.5
TQA*	79.0 ± 7.0	70.7 ± 16.6	88.5 ± 1.9	84.2 ± 7.9
PP				
Fixed Angle*	-	89.3 ± 1.1	86.2 ± 1.7	89.6 ± 1.5
Fixed Angle [†]	-	88.2 ± 1.5	84.2 ± 1.4	89.4 ± 1.6
Fourier*	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.2 ± 1.8
Interp.*	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.1 ± 1.5
Linear Ramp	91.4 ± 1.0	87.5 ± 0.7	84.4 ± 1.8	87.9 ± 0.7
Linear Ramp*	92.9 ± 1.1	88.9 ± 0.8	86.6 ± 2.0	88.6 ± 0.7
Recursive TS	-	-	-	-
TQA	83.7 ± 1.4	81.9 ± 1.3	75.7 ± 4.8	85.1 ± 1.8
TQA*	89.2 ± 1.9	86.8 ± 1.2	83.5 ± 1.9	87.8 ± 1.4
SV				
Fixed Angle*	91.4 ± 1.6	-	91.7 ± 2.4	93.4 ± 2.2
Fixed Angle [†]	89.0 ± 2.1	-	90.9 ± 2.4	92.8 ± 2.2
Fourier*	78.3 ± 10.6	81.3 ± 6.1	80.7 ± 7.8	90.3 ± 4.4
Interp.*	91.2 ± 1.6	91.3 ± 1.4	90.3 ± 3.8	91.0 ± 4.8
Linear Ramp	89.4 ± 1.4	88.7 ± 2.5	87.4 ± 2.1	90.4 ± 2.2
Recursive TS	90.1 ± 2.2	89.0 ± 2.8	87.6 ± 2.3	90.1 ± 3.5
TQA	84.8 ± 3.8	81.8 ± 4.3	79.9 ± 6.0	84.0 ± 5.6
TQA*	89.4 ± 1.9	88.8 ± 1.8	87.8 ± 2.7	91.0 ± 2.5

Table 11: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	-	88.8 ± 1.3	82.8 ± 6.9	88.6 ± 8.8	-	10 [39]	21 [100]	13 [40]
Fixed Angle†	-	87.9 ± 1.5	77.1 ± 8.3	85.4 ± 11.0	-	10 [39]	21 [100]	13 [40]
Fourier*	75.6 ± 4.5	80.9 ± 2.7	74.4 ± 8.1	76.6 ± 5.5	20 [40–50]	20 [39–144]	55 [40–100]	17 [40]
Interp.*	87.0 ± 2.6	85.7 ± 3.4	86.1 ± 6.2	77.7 ± 6.7	20 [40–50]	20 [39–144]	55 [40–100]	17 [40]
Linear Ramp	80.1 ± 4.6	79.9 ± 1.7	88.6 ± 1.9	72.8 ± 5.2	10 [50]	10 [144]	10 [100]	7 [10]
Linear Ramp*	85.8 ± 4.1	81.4 ± 1.0	91.1 ± 1.2	75.0 ± 6.0	10 [50]	10 [144]	10 [100]	7 [10]
TQA	69.6 ± 1.8	79.8 ± 2.1	74.4 ± 9.2	75.4 ± 5.1	20 [40–50]	20 [39–144]	56 [40–100]	17 [40]
TQA*	77.8 ± 4.9	82.5 ± 2.7	78.7 ± 9.6	79.2 ± 6.3	20 [40–50]	20 [39–144]	56 [40–100]	17 [40]
MPS (Quimb)								
Fixed Angle*	-	52.8 ± 1.1	90.5 ± 1.4	89.5 ± 5.9	-	10 [144]	30 [100]	13 [40]
Fixed Angle†	-	51.0 ± 0.9	90.4 ± 1.5	84.8 ± 10.7	-	10 [144]	30 [100]	13 [40]
Fourier*	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	83.6 ± 9.6	20 [40–50]	31 [39–144]	130 [40–100]	28 [40]
Interp.*	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	87.3 ± 8.4	20 [40–50]	31 [39–144]	130 [40–100]	28 [40]
Linear Ramp	90.5 ± 1.5	69.3 ± 2.0	87.7 ± 1.5	75.7 ± 5.4	10 [50]	10 [144]	10 [100]	7 [10]
Linear Ramp*	92.4 ± 1.9	68.8 ± 2.0	90.2 ± 1.0	77.8 ± 7.4	10 [50]	10 [144]	10 [100]	7 [10]
Recursive TS	-	-	-	-	-	-	-	-
TQA	70.5 ± 3.4	68.1 ± 14.3	84.2 ± 2.4	77.4 ± 5.5	20 [40–50]	20 [39–144]	130 [40–100]	27 [40]
TQA*	79.0 ± 7.0	70.7 ± 16.6	88.5 ± 1.9	84.2 ± 7.9	20 [40–50]	20 [39–144]	130 [40–100]	27 [40]
PP								
Fixed Angle*	-	89.3 ± 1.1	86.2 ± 1.7	89.6 ± 1.5	-	30 [39–144]	21 [100]	13 [40]
Fixed Angle†	-	88.2 ± 1.5	84.2 ± 1.4	89.4 ± 1.6	-	30 [39–144]	21 [100]	13 [40]
Fourier*	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.2 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	28 [40]
Interp.*	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.1 ± 1.5	20 [40–50]	31 [39–144]	130 [40–100]	28 [40]
Linear Ramp	91.4 ± 1.0	87.5 ± 0.7	84.4 ± 1.8	87.9 ± 0.7	10 [50]	20 [105–144]	20 [40–100]	8 [10]
Linear Ramp*	92.9 ± 1.1	88.9 ± 0.8	86.6 ± 2.0	88.6 ± 0.7	10 [50]	20 [105–144]	20 [40–100]	8 [10]
Recursive TS	-	-	-	-	-	-	-	-
TQA	83.7 ± 1.4	81.9 ± 1.3	75.7 ± 4.8	85.1 ± 1.8	20 [40–50]	20 [39–144]	130 [40–100]	27 [40]
TQA*	89.2 ± 1.9	86.8 ± 1.2	83.5 ± 1.9	87.8 ± 1.4	20 [40–50]	20 [39–144]	130 [40–100]	27 [40]
SV								
Fixed Angle*	91.4 ± 1.6	-	91.7 ± 2.4	93.4 ± 2.2	61 [10–20]	-	106 [10–20]	155 [10]
Fixed Angle†	89.0 ± 2.1	-	90.9 ± 2.4	92.8 ± 2.2	61 [10–20]	-	106 [10–20]	158 [10]
Fourier*	78.3 ± 10.6	81.3 ± 6.1	80.7 ± 7.8	90.3 ± 4.4	91 [10–20]	20 [12–21]	784 [10–20]	567 [10]
Interp.*	91.2 ± 1.6	91.3 ± 1.4	90.3 ± 3.8	91.0 ± 4.8	91 [10–20]	20 [12–21]	827 [10–22]	372 [10]
Linear Ramp	89.4 ± 1.4	88.7 ± 2.5	87.4 ± 2.1	90.4 ± 2.2	10 [20]	10 [21]	10 [20]	7 [2]
Recursive TS	90.1 ± 2.2	89.0 ± 2.8	87.6 ± 2.3	90.1 ± 3.5	90 [10–20]	20 [12–21]	430 [10–22]	353 [10]
TQA	84.8 ± 3.8	81.8 ± 4.3	79.9 ± 6.0	84.0 ± 5.6	92 [10–20]	20 [12–21]	821 [10–22]	355 [10]
TQA*	89.4 ± 1.9	88.8 ± 1.8	87.8 ± 2.7	91.0 ± 2.5	91 [10–20]	20 [12–21]	821 [10–22]	142 [10]

Table 12: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

5 Tables for depth $P = 5$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	10 [144]	2 [100]	2 [100]
Fixed Angle†	-	10 [144]	2 [100]	2 [100]
Fourier*	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]
Linear Ramp	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angle*	-	-	2 [100]	12 [40–100]
Fixed Angle†	-	-	2 [100]	12 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
TQA*	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
PP				
Fixed Angle*	-	10 [144]	2 [100]	12 [40–100]
Fixed Angle†	-	10 [144]	2 [100]	12 [40–100]
Fourier*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]
Linear Ramp	10 [50]	20 [105–144]	29 [40–100]	8 [100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
TQA*	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]
SV				
Fixed Angle*	46 [10–20]	-	60 [10–20]	115 [10–20]
Fixed Angle†	46 [10–20]	-	61 [10–20]	121 [10–20]
Fourier*	90 [10–20]	20 [12–21]	784 [10–20]	366 [10–20]
Interp.*	91 [10–20]	20 [12–21]	781 [10–20]	372 [10–20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	90 [10–20]	20 [12–21]	293 [10–20]	348 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	359 [10–20]
TQA*	90 [10–20]	20 [12–21]	672 [10–20]	142 [10–20]

Table 13: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	84.3 ± 3.3	94.5 ± 0.7	79.1 ± 2.5
Fixed Angle [†]	-	83.3 ± 3.4	94.4 ± 0.6	68.1 ± 0.6
Fourier*	79.9 ± 6.1	82.4 ± 3.3	78.1 ± 10.5	76.8 ± 6.3
Interp.*	90.5 ± 2.0	87.2 ± 3.8	89.3 ± 5.3	80.8 ± 9.1
Linear Ramp	84.4 ± 4.3	85.7 ± 1.0	78.4 ± 14.1	79.6 ± 5.4
Linear Ramp*	88.8 ± 2.7	82.0 ± 2.0	93.4 ± 0.8	82.3 ± 5.4
TQA	67.9 ± 7.7	79.2 ± 1.2	89.2 ± 1.8	74.9 ± 7.2
TQA*	75.4 ± 2.6	81.5 ± 1.3	91.0 ± 1.2	80.1 ± 8.9
MPS (Quimb)				
Fixed Angle*	-	-	94.4 ± 0.7	93.2 ± 4.2
Fixed Angle [†]	-	-	94.3 ± 0.7	90.0 ± 8.9
Fourier*	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	86.4 ± 10.4
Interp.*	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	90.4 ± 7.1
Linear Ramp	92.7 ± 1.2	68.4 ± 2.0	90.0 ± 1.5	84.5 ± 4.0
Linear Ramp*	94.2 ± 1.5	70.2 ± 1.3	92.4 ± 1.0	86.3 ± 4.9
Recursive TS	-	71.1	-	83.7 ± 7.3
TQA	66.5 ± 5.3	73.4 ± 4.6	87.8 ± 1.5	79.0 ± 7.8
TQA*	77.8 ± 6.5	75.2 ± 4.8	90.9 ± 1.2	85.4 ± 8.1
PP				
Fixed Angle*	-	90.5 ± 0.8	89.1 ± 0.1	91.4 ± 1.6
Fixed Angle [†]	-	88.7 ± 1.8	86.3 ± 0.9	91.1 ± 1.7
Fourier*	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	88.6 ± 1.6
Interp.*	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.3 ± 1.8
Linear Ramp	92.5 ± 1.0	89.0 ± 0.8	86.2 ± 1.8	88.9 ± 0.8
Linear Ramp*	94.0 ± 1.0	90.2 ± 0.6	88.1 ± 2.1	89.6 ± 0.8
Recursive TS	-	88.2	-	89.0 ± 2.3
TQA	83.6 ± 0.9	83.8 ± 1.1	72.8 ± 5.9	85.6 ± 1.5
TQA*	93.3 ± 1.4	88.2 ± 0.8	84.1 ± 2.0	89.2 ± 1.4
SV				
Fixed Angle*	93.1 ± 1.4	-	94.8 ± 2.1	95.3 ± 2.0
Fixed Angle [†]	90.1 ± 2.2	-	93.7 ± 2.1	94.9 ± 2.0
Fourier*	80.3 ± 10.2	79.6 ± 6.2	81.2 ± 8.2	85.2 ± 10.7
Interp.*	92.9 ± 1.4	92.2 ± 1.0	92.4 ± 3.9	92.5 ± 4.4
Linear Ramp	90.7 ± 1.3	90.8 ± 0.7	89.0 ± 1.9	91.5 ± 2.1
Recursive TS	91.7 ± 2.1	90.5 ± 2.5	89.7 ± 3.5	91.1 ± 3.5
TQA	86.9 ± 3.6	83.6 ± 4.3	81.9 ± 6.3	86.6 ± 5.3
TQA*	91.6 ± 1.5	90.2 ± 1.8	90.6 ± 3.3	93.1 ± 2.2

Table 14: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for MAXCUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	ER	Approximation Ratio			RR	Num. Instances			RR
		HH	LB			HH	LB		
MPS (Aer)									
Fixed Angle*	-	84.3 $\pm$ 3.3	94.5 $\pm$ 0.7	79.1 $\pm$ 2.5	-	10 [144]	2 [100]	2 [100]	2 [100]
Fixed Angle [†]	-	83.3 $\pm$ 3.4	94.4 $\pm$ 0.6	68.1 $\pm$ 0.6	-	10 [144]	2 [100]	2 [100]	2 [100]
Fourier*	79.9 $\pm$ 6.1	82.4 $\pm$ 3.3	78.1 $\pm$ 10.5	76.8 $\pm$ 6.3	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]	17 [40–100]
Interp.*	90.5 $\pm$ 2.0	87.2 $\pm$ 3.8	89.3 $\pm$ 5.3	80.8 $\pm$ 9.1	20 [40–50]	20 [39–144]	55 [40–100]	17 [40–100]	17 [40–100]
Linear Ramp	84.4 $\pm$ 4.3	85.7 $\pm$ 1.0	78.4 $\pm$ 14.1	79.6 $\pm$ 5.4	10 [50]	10 [144]	19 [100]	7 [100]	7 [100]
Linear Ramp*	88.8 $\pm$ 2.7	82.0 $\pm$ 2.0	93.4 $\pm$ 0.8	82.3 $\pm$ 5.4	10 [50]	10 [144]	10 [100]	7 [100]	7 [100]
TQA	67.9 $\pm$ 7.7	79.2 $\pm$ 1.2	89.2 $\pm$ 1.8	74.9 $\pm$ 7.2	10 [50]	10 [144]	10 [100]	7 [100]	7 [100]
TQA*	75.4 $\pm$ 2.6	81.5 $\pm$ 1.3	91.0 $\pm$ 1.2	80.1 $\pm$ 8.9	10 [50]	10 [144]	10 [100]	7 [100]	7 [100]
MPS (Quimb)									
Fixed Angle*	-	-	94.4 $\pm$ 0.7	93.2 $\pm$ 4.2	-	-	2 [100]	12 [40–100]	12 [40–100]
Fixed Angle [†]	-	-	94.3 $\pm$ 0.7	90.0 $\pm$ 8.9	-	-	2 [100]	12 [40–100]	12 [40–100]
Fourier*	82.0 $\pm$ 7.5	77.1 $\pm$ 8.9	85.8 $\pm$ 4.6	86.4 $\pm$ 10.4	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]	28 [40–100]
Interp.*	94.5 $\pm$ 1.2	81.5 $\pm$ 8.7	92.4 $\pm$ 1.1	90.4 $\pm$ 7.1	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]	28 [40–100]
Linear Ramp	92.7 $\pm$ 1.2	68.4 $\pm$ 2.0	90.0 $\pm$ 1.5	84.5 $\pm$ 4.0	10 [50]	10 [144]	10 [100]	7 [100]	7 [100]
Linear Ramp*	94.2 $\pm$ 1.5	70.2 $\pm$ 1.3	92.4 $\pm$ 1.0	86.3 $\pm$ 4.9	10 [50]	10 [144]	10 [100]	7 [100]	7 [100]
Recursive TS	-	71.1	-	83.7 $\pm$ 7.3	-	1 [106]	-	2 [40–100]	2 [40–100]
TQA	66.5 $\pm$ 5.3	73.4 $\pm$ 4.6	87.8 $\pm$ 1.5	79.0 $\pm$ 7.8	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]	17 [40–100]
TQA*	77.8 $\pm$ 6.5	75.2 $\pm$ 4.8	90.9 $\pm$ 1.2	85.4 $\pm$ 8.1	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]	17 [40–100]
PP									
Fixed Angle*	-	90.5 $\pm$ 0.8	89.1 $\pm$ 0.1	91.4 $\pm$ 1.6	-	10 [144]	2 [100]	12 [40–100]	12 [40–100]
Fixed Angle [†]	-	88.7 $\pm$ 1.8	86.3 $\pm$ 0.9	91.1 $\pm$ 1.7	-	10 [144]	2 [100]	12 [40–100]	12 [40–100]
Fourier*	89.6 $\pm$ 1.9	88.4 $\pm$ 2.7	83.5 $\pm$ 2.5	88.6 $\pm$ 1.6	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]	28 [40–100]
Interp.*	91.9 $\pm$ 2.0	91.1 $\pm$ 1.0	86.6 $\pm$ 1.8	90.3 $\pm$ 1.8	20 [40–50]	31 [39–144]	130 [40–100]	28 [40–100]	28 [40–100]
Linear Ramp	92.5 $\pm$ 1.0	89.0 $\pm$ 0.8	86.2 $\pm$ 1.8	88.9 $\pm$ 0.8	10 [50]	20 [105–144]	29 [40–100]	8 [100]	8 [100]
Linear Ramp*	94.0 $\pm$ 1.0	90.2 $\pm$ 0.6	88.1 $\pm$ 2.1	89.6 $\pm$ 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [100]	8 [100]
Recursive TS	-	88.2	-	89.0 $\pm$ 2.3	-	1 [106]	-	2 [40–100]	2 [40–100]
TQA	83.6 $\pm$ 0.9	83.8 $\pm$ 1.1	72.8 $\pm$ 5.9	85.6 $\pm$ 1.5	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]	17 [40–100]
TQA*	93.3 $\pm$ 1.4	88.2 $\pm$ 0.8	84.1 $\pm$ 2.0	89.2 $\pm$ 1.4	10 [50]	20 [105–144]	20 [40–100]	17 [40–100]	17 [40–100]
SV									
Fixed Angle*	93.1 $\pm$ 1.4	-	94.8 $\pm$ 2.1	95.3 $\pm$ 2.0	46 [10–20]	-	60 [10–20]	115 [10–20]	115 [10–20]
Fixed Angle [†]	90.1 $\pm$ 2.2	-	93.7 $\pm$ 2.1	94.9 $\pm$ 2.0	46 [10–20]	-	61 [10–20]	121 [10–20]	121 [10–20]
Fourier*	80.3 $\pm$ 10.2	79.6 $\pm$ 6.2	81.2 $\pm$ 8.2	85.2 $\pm$ 10.7	90 [10–20]	20 [12–21]	784 [10–20]	366 [10–20]	366 [10–20]
Interp.*	92.9 $\pm$ 1.4	92.2 $\pm$ 1.0	92.4 $\pm$ 3.9	92.5 $\pm$ 4.4	91 [10–20]	20 [12–21]	781 [10–20]	372 [10–20]	372 [10–20]
Linear Ramp	90.7 $\pm$ 1.3	90.8 $\pm$ 0.7	89.0 $\pm$ 1.9	91.5 $\pm$ 2.1	10 [20]	10 [21]	10 [20]	7 [20]	7 [20]
Recursive TS	91.7 $\pm$ 2.1	90.5 $\pm$ 2.5	89.7 $\pm$ 3.5	91.1 $\pm$ 3.5	90 [10–20]	20 [12–21]	293 [10–20]	348 [10–20]	348 [10–20]
TQA	86.9 $\pm$ 3.6	83.6 $\pm$ 4.3	81.9 $\pm$ 6.3	86.6 $\pm$ 5.3	92 [10–20]	20 [12–21]	677 [10–20]	359 [10–20]	359 [10–20]
TQA*	91.6 $\pm$ 1.5	90.2 $\pm$ 1.8	90.6 $\pm$ 3.3	93.1 $\pm$ 2.2	90 [10–20]	20 [12–21]	672 [10–20]	142 [10–20]	142 [10–20]

Table 15: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

6 Tables for depth $P = 6$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	10 [39]	20 [100]	11 [40-100]
Fixed Angle†	-	10 [39]	20 [100]	11 [40-100]
Fourier*	20 [40-50]	20 [39-144]	55 [40-100]	17 [40-100]
Interp.*	20 [40-50]	20 [39-144]	55 [40-100]	17 [40-100]
Linear Ramp	10 [50]	10 [144]	19 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	20 [40-50]	20 [39-144]	56 [40-100]	17 [40-100]
TQA*	20 [40-50]	20 [39-144]	56 [40-100]	17 [40-100]
MPS (Quimb)				
Fixed Angle*	-	10 [144]	10 [100]	11 [40-100]
Fixed Angle†	-	10 [144]	10 [100]	11 [40-100]
Fourier*	20 [40-50]	31 [39-144]	130 [40-100]	28 [40-100]
Interp.*	20 [40-50]	31 [39-144]	130 [40-100]	28 [40-100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-
TQA	20 [40-50]	20 [39-144]	130 [40-100]	27 [40-100]
TQA*	20 [40-50]	20 [39-144]	130 [40-100]	27 [40-100]
PP				
Fixed Angle*	-	30 [39-144]	10 [100]	11 [40-100]
Fixed Angle†	-	30 [39-144]	10 [100]	11 [40-100]
Fourier*	20 [40-50]	31 [39-144]	130 [40-100]	28 [40-100]
Interp.*	20 [40-50]	31 [39-144]	130 [40-100]	28 [40-100]
Linear Ramp	10 [50]	20 [105-144]	29 [40-100]	8 [100]
Linear Ramp*	10 [50]	20 [105-144]	20 [40-100]	8 [100]
Recursive TS	-	-	-	-
TQA	20 [40-50]	20 [39-144]	130 [40-100]	27 [40-100]
TQA*	20 [40-50]	20 [39-144]	130 [40-100]	27 [40-100]
SV				
Fixed Angle*	17 [11-20]	-	20 [10-20]	59 [10-20]
Fixed Angle†	17 [11-20]	-	20 [10-20]	59 [10-20]
Fourier*	90 [10-20]	20 [12-21]	781 [10-20]	366 [10-20]
Interp.*	91 [10-20]	20 [12-21]	781 [10-20]	372 [10-20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	89 [10-20]	20 [12-21]	286 [10-20]	348 [10-20]
TQA	92 [10-20]	20 [12-21]	677 [10-20]	360 [10-20]
TQA*	90 [10-20]	20 [12-21]	672 [10-20]	142 [10-20]

Table 16: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	91.6 ± 0.9	90.1 ± 2.9	95.6 ± 5.3
Fixed Angle†	-	89.9 ± 1.7	86.0 ± 5.5	94.3 ± 7.6
Fourier*	78.1 ± 5.7	83.1 ± 3.0	80.1 ± 9.2	81.4 ± 7.6
Interp.*	92.2 ± 1.8	88.3 ± 4.4	91.8 ± 3.8	84.7 ± 7.0
Linear Ramp	88.8 ± 2.3	81.3 ± 2.8	79.6 ± 14.0	83.1 ± 3.4
Linear Ramp*	92.0 ± 1.8	81.4 ± 1.7	94.6 ± 0.9	86.4 ± 2.2
TQA	74.3 ± 4.5	82.9 ± 2.5	80.2 ± 8.6	82.2 ± 7.4
TQA*	80.9 ± 3.8	84.6 ± 3.1	84.5 ± 8.8	87.8 ± 8.6
MPS (Quimb)				
Fixed Angle*	-	53.9 ± 2.0	95.1 ± 1.1	96.1 ± 2.4
Fixed Angle†	-	50.9 ± 1.1	95.1 ± 1.0	94.7 ± 4.5
Fourier*	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	88.9 ± 9.6
Interp.*	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	92.0 ± 6.2
Linear Ramp	94.0 ± 1.3	67.6 ± 1.8	91.4 ± 1.3	83.8 ± 8.7
Linear Ramp*	95.2 ± 1.6	69.2 ± 1.6	93.9 ± 1.3	86.4 ± 7.3
Recursive TS	-	-	-	-
TQA	71.7 ± 6.0	70.0 ± 16.1	88.0 ± 2.1	85.9 ± 8.9
TQA*	85.5 ± 6.7	71.7 ± 17.6	91.1 ± 1.4	90.6 ± 7.1
PP				
Fixed Angle*	-	92.0 ± 0.7	89.1 ± 1.8	92.3 ± 1.6
Fixed Angle†	-	90.2 ± 1.5	85.5 ± 1.6	91.9 ± 1.7
Fourier*	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.5 ± 1.8
Interp.*	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.1 ± 2.0
Linear Ramp	93.2 ± 1.0	90.0 ± 0.7	87.2 ± 1.8	89.5 ± 0.8
Linear Ramp*	94.7 ± 1.0	91.3 ± 0.7	89.2 ± 2.2	90.4 ± 0.7
Recursive TS	-	-	-	-
TQA	84.6 ± 1.9	85.6 ± 1.2	76.5 ± 5.9	87.5 ± 2.1
TQA*	91.3 ± 2.4	89.6 ± 0.9	85.2 ± 2.2	90.3 ± 1.6
SV				
Fixed Angle*	94.4 ± 1.5	-	96.5 ± 1.4	96.2 ± 1.4
Fixed Angle†	90.3 ± 2.5	-	95.6 ± 1.3	95.9 ± 1.3
Fourier*	80.2 ± 10.7	78.5 ± 9.3	80.9 ± 9.2	84.8 ± 11.7
Interp.*	94.1 ± 1.3	93.2 ± 1.2	93.7 ± 3.8	93.1 ± 4.9
Linear Ramp	91.8 ± 1.3	91.4 ± 1.3	90.1 ± 1.7	92.4 ± 1.9
Recursive TS	92.9 ± 1.9	91.2 ± 2.4	90.9 ± 2.7	92.2 ± 3.4
TQA	88.3 ± 3.3	85.4 ± 3.9	82.5 ± 7.3	88.4 ± 5.0
TQA*	92.9 ± 1.4	90.8 ± 1.9	91.9 ± 3.0	94.2 ± 2.1

Table 17: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	-	91.6 ± 0.9	90.1 ± 2.9	95.6 ± 5.3	-	10 [39]	20 [100]	11 [40]
Fixed Angle†	-	89.9 ± 1.7	86.0 ± 5.5	94.3 ± 7.6	-	10 [39]	20 [100]	11 [40]
Fourier*	78.1 ± 5.7	83.1 ± 3.0	80.1 ± 9.2	81.4 ± 7.6	20 [40–50]	20 [39–144]	55 [40–100]	17 [40]
Interp.*	92.2 ± 1.8	88.3 ± 4.4	91.8 ± 3.8	84.7 ± 7.0	20 [40–50]	20 [39–144]	55 [40–100]	17 [40]
Linear Ramp	88.8 ± 2.3	81.3 ± 2.8	79.6 ± 14.0	83.1 ± 3.4	10 [50]	10 [144]	19 [100]	7 [40]
Linear Ramp*	92.0 ± 1.8	81.4 ± 1.7	94.6 ± 0.9	86.4 ± 2.2	10 [50]	10 [144]	10 [100]	7 [40]
TQA	74.3 ± 4.5	82.9 ± 2.5	80.2 ± 8.6	82.2 ± 7.4	20 [40–50]	20 [39–144]	56 [40–100]	17 [40]
TQA*	80.9 ± 3.8	84.6 ± 3.1	84.5 ± 8.8	87.8 ± 8.6	20 [40–50]	20 [39–144]	56 [40–100]	17 [40]
MPS (Quimb)								
Fixed Angle*	-	53.9 ± 2.0	95.1 ± 1.1	96.1 ± 2.4	-	10 [144]	10 [100]	11 [40]
Fixed Angle†	-	50.9 ± 1.1	95.1 ± 1.0	94.7 ± 4.5	-	10 [144]	10 [100]	11 [40]
Fourier*	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	88.9 ± 9.6	20 [40–50]	31 [39–144]	130 [40–100]	28 [40]
Interp.*	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	92.0 ± 6.2	20 [40–50]	31 [39–144]	130 [40–100]	28 [40]
Linear Ramp	94.0 ± 1.3	67.6 ± 1.8	91.4 ± 1.3	83.8 ± 8.7	10 [50]	10 [144]	10 [100]	7 [40]
Linear Ramp*	95.2 ± 1.6	69.2 ± 1.6	93.9 ± 1.3	86.4 ± 7.3	10 [50]	10 [144]	10 [100]	7 [40]
Recursive TS	-	-	-	-	-	-	-	-
TQA	71.7 ± 6.0	70.0 ± 16.1	88.0 ± 2.1	85.9 ± 8.9	20 [40–50]	20 [39–144]	130 [40–100]	27 [40]
TQA*	85.5 ± 6.7	71.7 ± 17.6	91.1 ± 1.4	90.6 ± 7.1	20 [40–50]	20 [39–144]	130 [40–100]	27 [40]
PP								
Fixed Angle*	-	92.0 ± 0.7	89.1 ± 1.8	92.3 ± 1.6	-	30 [39–144]	10 [100]	11 [40]
Fixed Angle†	-	90.2 ± 1.5	85.5 ± 1.6	91.9 ± 1.7	-	30 [39–144]	10 [100]	11 [40]
Fourier*	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.5 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	28 [40]
Interp.*	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.1 ± 2.0	20 [40–50]	31 [39–144]	130 [40–100]	28 [40]
Linear Ramp	93.2 ± 1.0	90.0 ± 0.7	87.2 ± 1.8	89.5 ± 0.8	10 [50]	20 [105–144]	29 [40–100]	8 [40]
Linear Ramp*	94.7 ± 1.0	91.3 ± 0.7	89.2 ± 2.2	90.4 ± 0.7	10 [50]	20 [105–144]	20 [40–100]	8 [40]
Recursive TS	-	-	-	-	-	-	-	-
TQA	84.6 ± 1.9	85.6 ± 1.2	76.5 ± 5.9	87.5 ± 2.1	20 [40–50]	20 [39–144]	130 [40–100]	27 [40]
TQA*	91.3 ± 2.4	89.6 ± 0.9	85.2 ± 2.2	90.3 ± 1.6	20 [40–50]	20 [39–144]	130 [40–100]	27 [40]
SV								
Fixed Angle*	94.4 ± 1.5	-	96.5 ± 1.4	96.2 ± 1.4	17 [11–20]	-	20 [10–20]	59 [10]
Fixed Angle†	90.3 ± 2.5	-	95.6 ± 1.3	95.9 ± 1.3	17 [11–20]	-	20 [10–20]	59 [10]
Fourier*	80.2 ± 10.7	78.5 ± 9.3	80.9 ± 9.2	84.8 ± 11.7	90 [10–20]	20 [12–21]	781 [10–20]	366 [10]
Interp.*	94.1 ± 1.3	93.2 ± 1.2	93.7 ± 3.8	93.1 ± 4.9	91 [10–20]	20 [12–21]	781 [10–20]	372 [10]
Linear Ramp	91.8 ± 1.3	91.4 ± 1.3	90.1 ± 1.7	92.4 ± 1.9	10 [20]	10 [21]	10 [20]	7 [10]
Recursive TS	92.9 ± 1.9	91.2 ± 2.4	90.9 ± 2.7	92.2 ± 3.4	89 [10–20]	20 [12–21]	286 [10–20]	348 [10]
TQA	88.3 ± 3.3	85.4 ± 3.9	82.5 ± 7.3	88.4 ± 5.0	92 [10–20]	20 [12–21]	677 [10–20]	360 [10]
TQA*	92.9 ± 1.4	90.8 ± 1.9	91.9 ± 3.0	94.2 ± 2.1	90 [10–20]	20 [12–21]	672 [10–20]	142 [10]

Table 18: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

7 Tables for depth $P = 7$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	1 [100]	1 [100]
Fixed Angle†	-	-	1 [100]	1 [100]
Fourier*	20 [40–50]	20 [39–144]	49 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	49 [40–100]	17 [40–100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angle*	-	10 [144]	1 [100]	1 [100]
Fixed Angle†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
PP				
Fixed Angle*	-	10 [144]	1 [100]	1 [100]
Fixed Angle†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	-	-	-	-
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
SV				
Fixed Angle*	17 [11–20]	-	19 [10–20]	59 [10–20]
Fixed Angle†	17 [11–20]	-	19 [10–20]	60 [10–20]
Fourier*	90 [10–20]	20 [12–21]	780 [10–20]	366 [10–20]
Interp.*	91 [10–20]	20 [12–21]	780 [10–20]	372 [10–20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	89 [10–20]	20 [12–21]	287 [10–20]	349 [10–20]
TQA	92 [10–20]	20 [12–21]	677 [10–20]	361 [10–20]
TQA*	90 [10–20]	20 [12–21]	672 [10–20]	141 [10–20]

Table 19: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	97.0	82.6
Fixed Angle [†]	-	-	96.8	71.9
Fourier*	79.4 ± 5.7	84.4 ± 3.2	80.3 ± 9.4	83.7 ± 8.2
Interp.*	93.7 ± 2.1	89.2 ± 4.7	93.4 ± 3.0	85.0 ± 8.3
Linear Ramp	89.2 ± 3.2	81.9 ± 1.4	93.3 ± 1.0	85.4 ± 1.9
Linear Ramp*	92.9 ± 2.0	83.5 ± 2.5	95.2 ± 0.8	87.1 ± 3.3
TQA	78.1 ± 5.3	82.0 ± 1.5	90.7 ± 1.1	79.2 ± 7.8
TQA*	81.7 ± 2.5	83.2 ± 1.9	92.5 ± 1.2	86.8 ± 4.5
MPS (Quimb)				
Fixed Angle*	-	56.3 ± 2.6	97.0	91.9
Fixed Angle [†]	-	52.3 ± 2.6	96.7	84.8
Fourier*	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	89.0 ± 8.8
Interp.*	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	93.0 ± 6.0
Linear Ramp	94.7 ± 1.3	68.3 ± 3.0	92.5 ± 1.3	87.6 ± 4.1
Linear Ramp*	95.9 ± 1.4	71.7 ± 1.7	94.6 ± 0.8	90.2 ± 2.5
Recursive TS	-	-	-	-
TQA	71.9 ± 9.7	57.3 ± 1.0	89.6 ± 1.0	76.8 ± 11.9
TQA*	86.8 ± 4.8	57.5 ± 1.0	92.4 ± 0.8	83.2 ± 8.5
PP				
Fixed Angle*	-	92.5 ± 0.6	90.6	90.7
Fixed Angle [†]	-	90.2 ± 1.6	86.4	89.8
Fourier*	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	89.9 ± 1.7
Interp.*	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.2 ± 2.1
Linear Ramp	93.8 ± 1.0	90.9 ± 0.6	87.6 ± 1.9	90.0 ± 0.8
Linear Ramp*	95.1 ± 1.0	92.1 ± 0.6	89.8 ± 2.3	90.8 ± 0.8
Recursive TS	-	-	-	-
TQA	83.8 ± 0.9	87.0 ± 0.8	77.2 ± 6.9	86.3 ± 2.3
TQA*	94.5 ± 1.5	90.2 ± 0.6	85.6 ± 2.3	89.6 ± 1.1
SV				
Fixed Angle*	95.4 ± 1.5	-	97.4 ± 1.2	97.2 ± 1.2
Fixed Angle [†]	90.7 ± 3.0	-	96.3 ± 1.0	96.8 ± 1.2
Fourier*	80.3 ± 10.8	77.9 ± 10.7	80.8 ± 9.7	84.4 ± 12.4
Interp.*	95.1 ± 1.2	93.9 ± 1.2	94.5 ± 4.0	93.3 ± 5.4
Linear Ramp	92.5 ± 1.2	92.0 ± 1.3	91.1 ± 1.6	93.1 ± 1.8
Recursive TS	93.8 ± 1.9	91.8 ± 2.3	92.3 ± 4.0	92.9 ± 3.4
TQA	89.4 ± 3.3	86.8 ± 3.7	83.1 ± 7.8	89.3 ± 4.8
TQA*	93.8 ± 1.3	91.4 ± 1.9	92.5 ± 3.2	95.0 ± 1.8

Table 20: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	-	-	97.0	82.6	-	-	1 [100]	1 [100]
Fixed Angle†	-	-	96.8	71.9	-	-	1 [100]	1 [100]
Fourier*	79.4 ± 5.7	84.4 ± 3.2	80.3 ± 9.4	83.7 ± 8.2	20 [40–50]	20 [39–144]	49 [40–100]	17 [40–100]
Interp.*	93.7 ± 2.1	89.2 ± 4.7	93.4 ± 3.0	85.0 ± 8.3	20 [40–50]	20 [39–144]	49 [40–100]	17 [40–100]
Linear Ramp	89.2 ± 3.2	81.9 ± 1.4	93.3 ± 1.0	85.4 ± 1.9	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	92.9 ± 2.0	83.5 ± 2.5	95.2 ± 0.8	87.1 ± 3.3	10 [50]	10 [144]	10 [100]	7 [100]
TQA	78.1 ± 5.3	82.0 ± 1.5	90.7 ± 1.1	79.2 ± 7.8	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	81.7 ± 2.5	83.2 ± 1.9	92.5 ± 1.2	86.8 ± 4.5	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)								
Fixed Angle*	-	56.3 ± 2.6	97.0	91.9	-	10 [144]	1 [100]	1 [100]
Fixed Angle†	-	52.3 ± 2.6	96.7	84.8	-	10 [144]	1 [100]	1 [100]
Fourier*	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	89.0 ± 8.8	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	93.0 ± 6.0	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp	94.7 ± 1.3	68.3 ± 3.0	92.5 ± 1.3	87.6 ± 4.1	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	95.9 ± 1.4	71.7 ± 1.7	94.6 ± 0.8	90.2 ± 2.5	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-	-	-	-	-
TQA	71.9 ± 9.7	57.3 ± 1.0	89.6 ± 1.0	76.8 ± 11.9	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	86.8 ± 4.8	57.5 ± 1.0	92.4 ± 0.8	83.2 ± 8.5	10 [50]	10 [144]	10 [100]	7 [100]
PP								
Fixed Angle*	-	92.5 ± 0.6	90.6	90.7	-	10 [144]	1 [100]	1 [100]
Fixed Angle†	-	90.2 ± 1.6	86.4	89.8	-	10 [144]	1 [100]	1 [100]
Fourier*	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	89.9 ± 1.7	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Interp.*	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.2 ± 2.1	20 [40–50]	31 [39–144]	120 [40–100]	28 [40–100]
Linear Ramp	93.8 ± 1.0	90.9 ± 0.6	87.6 ± 1.9	90.0 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp*	95.1 ± 1.0	92.1 ± 0.6	89.8 ± 2.3	90.8 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	-	-	-	-	-	-	-	-
TQA	83.8 ± 0.9	87.0 ± 0.8	77.2 ± 6.9	86.3 ± 2.3	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	94.5 ± 1.5	90.2 ± 0.6	85.6 ± 2.3	89.6 ± 1.1	10 [50]	10 [144]	10 [100]	7 [100]
SV								
Fixed Angle*	95.4 ± 1.5	-	97.4 ± 1.2	97.2 ± 1.2	17 [11–20]	-	19 [10–20]	59 [10–20]
Fixed Angle†	90.7 ± 3.0	-	96.3 ± 1.0	96.8 ± 1.2	17 [11–20]	-	19 [10–20]	60 [10–20]
Fourier*	80.3 ± 10.8	77.9 ± 10.7	80.8 ± 9.7	84.4 ± 12.4	90 [10–20]	20 [12–21]	780 [10–20]	366 [10–20]
Interp.*	95.1 ± 1.2	93.9 ± 1.2	94.5 ± 4.0	93.3 ± 5.4	91 [10–20]	20 [12–21]	780 [10–20]	372 [10–20]
Linear Ramp	92.5 ± 1.2	92.0 ± 1.3	91.1 ± 1.6	93.1 ± 1.8	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	93.8 ± 1.9	91.8 ± 2.3	92.3 ± 4.0	92.9 ± 3.4	89 [10–20]	20 [12–21]	287 [10–20]	349 [10–20]
TQA	89.4 ± 3.3	86.8 ± 3.7	83.1 ± 7.8	89.3 ± 4.8	92 [10–20]	20 [12–21]	677 [10–20]	361 [10–20]
TQA*	93.8 ± 1.3	91.4 ± 1.9	92.5 ± 3.2	95.0 ± 1.8	90 [10–20]	20 [12–21]	672 [10–20]	141 [10–20]

Table 21: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

8 Tables for depth $P = 8$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	10 [39]	11 [100]	11 [40-100]
Fixed Angle†	-	10 [39]	11 [100]	11 [40-100]
Fourier*	20 [40-50]	20 [39-144]	49 [40-100]	17 [40-100]
Interp.*	20 [40-50]	20 [39-144]	49 [40-100]	17 [40-100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	20 [40-50]	20 [39-144]	50 [40-100]	17 [40-100]
TQA*	20 [40-50]	20 [39-144]	50 [40-100]	17 [40-100]
MPS (Quimb)				
Fixed Angle*	-	10 [144]	10 [100]	11 [40-100]
Fixed Angle†	-	10 [144]	10 [100]	11 [40-100]
Fourier*	20 [40-50]	31 [39-144]	120 [40-100]	28 [40-100]
Interp.*	20 [40-50]	31 [39-144]	120 [40-100]	28 [40-100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-
TQA	20 [40-50]	20 [39-144]	120 [40-100]	27 [40-100]
TQA*	20 [40-50]	20 [39-144]	120 [40-100]	27 [40-100]
PP				
Fixed Angle*	-	30 [39-144]	10 [100]	11 [40-100]
Fixed Angle†	-	30 [39-144]	10 [100]	11 [40-100]
Fourier*	20 [40-50]	31 [39-144]	120 [40-100]	28 [40-100]
Interp.*	20 [40-50]	31 [39-144]	120 [40-100]	28 [40-100]
Linear Ramp	10 [50]	20 [105-144]	20 [40-100]	8 [100]
Linear Ramp*	10 [50]	20 [105-144]	20 [40-100]	8 [100]
Recursive TS	-	-	-	-
TQA	20 [40-50]	20 [39-144]	120 [40-100]	27 [40-100]
TQA*	20 [40-50]	20 [39-144]	120 [40-100]	27 [40-100]
SV				
Fixed Angle*	17 [11-20]	-	19 [10-20]	50 [10-20]
Fixed Angle†	17 [11-20]	-	19 [10-20]	50 [10-20]
Fourier*	90 [10-20]	20 [12-21]	778 [10-20]	366 [10-20]
Interp.*	91 [10-20]	20 [12-21]	779 [10-20]	372 [10-20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	89 [10-20]	20 [12-21]	282 [10-20]	349 [10-20]
TQA	92 [10-20]	20 [12-21]	677 [10-20]	356 [10-20]
TQA*	90 [10-20]	20 [12-21]	672 [10-20]	142 [10-20]

Table 22: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	92.5 ± 0.9	94.0 ± 1.4	97.2 ± 3.9
Fixed Angle†	-	90.8 ± 1.8	90.2 ± 2.9	96.8 ± 4.3
Fourier*	77.9 ± 5.5	85.2 ± 3.2	82.3 ± 8.7	81.8 ± 9.6
Interp.*	94.2 ± 2.1	90.0 ± 4.7	94.6 ± 2.6	86.6 ± 6.8
Linear Ramp	91.9 ± 1.7	83.5 ± 2.9	93.9 ± 0.9	87.3 ± 2.3
Linear Ramp*	94.5 ± 1.8	84.2 ± 1.4	95.6 ± 0.7	89.2 ± 2.4
TQA	81.5 ± 3.5	85.3 ± 2.7	83.8 ± 7.1	87.3 ± 6.6
TQA*	85.8 ± 2.2	86.5 ± 3.2	88.7 ± 5.9	92.2 ± 6.2
MPS (Quimb)				
Fixed Angle*	-	56.5 ± 3.5	96.7 ± 0.8	97.9 ± 1.3
Fixed Angle†	-	51.9 ± 2.6	96.6 ± 0.8	96.9 ± 3.3
Fourier*	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	91.1 ± 8.2
Interp.*	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	93.9 ± 6.5
Linear Ramp	95.3 ± 1.3	69.2 ± 1.8	93.4 ± 1.2	88.7 ± 2.4
Linear Ramp*	96.7 ± 0.8	69.0 ± 2.7	94.9 ± 1.2	93.8 ± 0.8
Recursive TS	-	-	-	-
TQA	82.2 ± 6.9	71.3 ± 17.4	89.3 ± 2.0	90.1 ± 8.0
TQA*	91.1 ± 3.7	72.6 ± 18.2	92.2 ± 1.3	93.4 ± 6.7
PP				
Fixed Angle*	-	93.4 ± 0.6	90.3 ± 1.9	93.2 ± 1.6
Fixed Angle†	-	91.0 ± 1.4	86.2 ± 1.7	92.6 ± 1.7
Fourier*	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.1 ± 2.1
Interp.*	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.5 ± 2.2
Linear Ramp	94.1 ± 1.0	91.6 ± 0.6	88.2 ± 1.9	90.4 ± 0.8
Linear Ramp*	95.5 ± 1.0	92.8 ± 0.6	90.3 ± 2.2	91.0 ± 0.8
Recursive TS	-	-	-	-
TQA	84.6 ± 2.2	88.2 ± 1.0	77.4 ± 6.7	89.1 ± 2.4
TQA*	91.7 ± 2.3	90.8 ± 0.7	85.6 ± 2.5	91.5 ± 1.5
SV				
Fixed Angle*	96.3 ± 1.4	-	98.0 ± 1.0	97.8 ± 1.1
Fixed Angle†	91.5 ± 3.3	-	96.7 ± 1.1	97.2 ± 1.1
Fourier*	78.8 ± 12.3	76.0 ± 10.2	79.7 ± 10.9	83.9 ± 13.0
Interp.*	95.7 ± 1.2	94.3 ± 1.1	95.2 ± 3.8	93.7 ± 5.4
Linear Ramp	93.2 ± 1.2	93.1 ± 0.7	92.0 ± 1.7	93.8 ± 1.7
Recursive TS	94.4 ± 1.8	92.2 ± 2.4	93.2 ± 4.5	93.4 ± 3.3
TQA	90.3 ± 3.4	88.0 ± 3.6	83.6 ± 8.5	90.0 ± 4.9
TQA*	94.4 ± 1.3	91.9 ± 1.7	93.1 ± 3.3	95.6 ± 1.7

Table 23: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	ER	Approximation Ratio				Num. Instances			
		HH	LB	RR		HH	LB		RR
MPS (Aer)									
Fixed Angle*	-	92.5 ± 0.9	94.0 ± 1.4	97.2 ± 3.9	-	10 [39]	11 [100]	11 [40]	11 [40]
Fixed Angle†	-	90.8 ± 1.8	90.2 ± 2.9	96.8 ± 4.3	-	10 [39]	11 [100]	11 [40]	11 [40]
Fourier*	77.9 ± 5.5	85.2 ± 3.2	82.3 ± 8.7	81.8 ± 9.6	20 [40–50]	20 [39–144]	49 [40–100]	17 [40]	17 [40]
Interp.*	94.2 ± 2.1	90.0 ± 4.7	94.6 ± 2.6	86.6 ± 6.8	20 [40–50]	20 [39–144]	49 [40–100]	17 [40]	17 [40]
Linear Ramp	91.9 ± 1.7	83.5 ± 2.9	93.9 ± 0.9	87.3 ± 2.3	10 [50]	10 [144]	10 [100]	7 [40]	7 [40]
Linear Ramp*	94.5 ± 1.8	84.2 ± 1.4	95.6 ± 0.7	89.2 ± 2.4	10 [50]	10 [144]	10 [100]	7 [40]	7 [40]
TQA	81.5 ± 3.5	85.3 ± 2.7	83.8 ± 7.1	87.3 ± 6.6	20 [40–50]	20 [39–144]	50 [40–100]	17 [40]	17 [40]
TQA*	85.8 ± 2.2	86.5 ± 3.2	88.7 ± 5.9	92.2 ± 6.2	20 [40–50]	20 [39–144]	50 [40–100]	17 [40]	17 [40]
MPS (Quimb)									
Fixed Angle*	-	56.5 ± 3.5	96.7 ± 0.8	97.9 ± 1.3	-	10 [144]	10 [100]	11 [40]	11 [40]
Fixed Angle†	-	51.9 ± 2.6	96.6 ± 0.8	96.9 ± 3.3	-	10 [144]	10 [100]	11 [40]	11 [40]
Fourier*	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	91.1 ± 8.2	20 [40–50]	31 [39–144]	120 [40–100]	28 [40]	28 [40]
Interp.*	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	93.9 ± 6.5	20 [40–50]	31 [39–144]	120 [40–100]	28 [40]	28 [40]
Linear Ramp	95.3 ± 1.3	69.2 ± 1.8	93.4 ± 1.2	88.7 ± 2.4	10 [50]	10 [144]	10 [100]	7 [40]	7 [40]
Linear Ramp*	96.7 ± 0.8	69.0 ± 2.7	94.9 ± 1.2	93.8 ± 0.8	10 [50]	10 [144]	10 [100]	7 [40]	7 [40]
Recursive TS	-	-	-	-	-	-	-	-	-
TQA	82.2 ± 6.9	71.3 ± 17.4	89.3 ± 2.0	90.1 ± 8.0	20 [40–50]	20 [39–144]	120 [40–100]	27 [40]	27 [40]
TQA*	91.1 ± 3.7	72.6 ± 18.2	92.2 ± 1.3	93.4 ± 6.7	20 [40–50]	20 [39–144]	120 [40–100]	27 [40]	27 [40]
PP									
Fixed Angle*	-	93.4 ± 0.6	90.3 ± 1.9	93.2 ± 1.6	-	30 [39–144]	10 [100]	11 [40]	11 [40]
Fixed Angle†	-	91.0 ± 1.4	86.2 ± 1.7	92.6 ± 1.7	-	30 [39–144]	10 [100]	11 [40]	11 [40]
Fourier*	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.1 ± 2.1	20 [40–50]	31 [39–144]	120 [40–100]	28 [40]	28 [40]
Interp.*	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.5 ± 2.2	20 [40–50]	31 [39–144]	120 [40–100]	28 [40]	28 [40]
Linear Ramp	94.1 ± 1.0	91.6 ± 0.6	88.2 ± 1.9	90.4 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [40]	8 [40]
Linear Ramp*	95.5 ± 1.0	92.8 ± 0.6	90.3 ± 2.2	91.0 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [40]	8 [40]
Recursive TS	-	-	-	-	-	-	-	-	-
TQA	84.6 ± 2.2	88.2 ± 1.0	77.4 ± 6.7	89.1 ± 2.4	20 [40–50]	20 [39–144]	120 [40–100]	27 [40]	27 [40]
TQA*	91.7 ± 2.3	90.8 ± 0.7	85.6 ± 2.5	91.5 ± 1.5	20 [40–50]	20 [39–144]	120 [40–100]	27 [40]	27 [40]
SV									
Fixed Angle*	96.3 ± 1.4	-	98.0 ± 1.0	97.8 ± 1.1	17 [11–20]	-	19 [10–20]	50 [10–20]	50 [10–20]
Fixed Angle†	91.5 ± 3.3	-	96.7 ± 1.1	97.2 ± 1.1	17 [11–20]	-	19 [10–20]	50 [10–20]	50 [10–20]
Fourier*	78.8 ± 12.3	76.0 ± 10.2	79.7 ± 10.9	83.9 ± 13.0	90 [10–20]	20 [12–21]	778 [10–20]	366 [10–20]	366 [10–20]
Interp.*	95.7 ± 1.2	94.3 ± 1.1	95.2 ± 3.8	93.7 ± 5.4	91 [10–20]	20 [12–21]	779 [10–20]	372 [10–20]	372 [10–20]
Linear Ramp	93.2 ± 1.2	93.1 ± 0.7	92.0 ± 1.7	93.8 ± 1.7	10 [20]	10 [21]	10 [20]	7 [20]	7 [20]
Recursive TS	94.4 ± 1.8	92.2 ± 2.4	93.2 ± 4.5	93.4 ± 3.3	89 [10–20]	20 [12–21]	282 [10–20]	349 [10–20]	349 [10–20]
TQA	90.3 ± 3.4	88.0 ± 3.6	83.6 ± 8.5	90.0 ± 4.9	92 [10–20]	20 [12–21]	677 [10–20]	356 [10–20]	356 [10–20]
TQA*	94.4 ± 1.3	91.9 ± 1.7	93.1 ± 3.3	95.6 ± 1.7	90 [10–20]	20 [12–21]	672 [10–20]	142 [10–20]	142 [10–20]

Table 24: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

9 Tables for depth $P = 9$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	1 [100]	1 [100]
Fixed Angle†	-	-	1 [100]	1 [100]
Fourier*	20 [40–50]	20 [39–144]	49 [40–100]	17 [40–100]
Interp.*	20 [40–50]	20 [39–144]	49 [40–100]	17 [40–100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)				
Fixed Angle*	-	10 [144]	1 [100]	1 [100]
Fixed Angle†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Interp.*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
PP				
Fixed Angle*	-	10 [144]	1 [100]	1 [100]
Fixed Angle†	-	10 [144]	1 [100]	1 [100]
Fourier*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Interp.*	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	-	-	-	-
TQA	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	10 [50]	10 [144]	10 [100]	7 [100]
SV				
Fixed Angle*	17 [11–20]	-	19 [10–20]	50 [10–20]
Fixed Angle†	17 [11–20]	-	19 [10–20]	50 [10–20]
Fourier*	90 [10–20]	20 [12–21]	776 [10–20]	364 [10–20]
Interp.*	91 [10–20]	20 [12–21]	777 [10–20]	371 [10–20]
Linear Ramp	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	89 [10–20]	20 [12–21]	281 [10–20]	352 [10–20]
TQA	92 [10–20]	20 [12–21]	676 [10–20]	356 [10–20]
TQA*	90 [10–20]	20 [12–21]	671 [10–20]	141 [10–20]

Table 25: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	97.8	90.2
Fixed Angle†	-	-	97.5	85.1
Fourier*	78.1 ± 5.3	85.8 ± 4.0	79.4 ± 8.3	84.0 ± 9.3
Interp.*	94.3 ± 3.6	90.6 ± 4.8	94.7 ± 2.5	88.7 ± 7.1
Linear Ramp	92.8 ± 1.5	84.0 ± 2.7	94.6 ± 1.0	87.8 ± 2.7
Linear Ramp*	95.4 ± 1.8	84.2 ± 2.0	96.0 ± 0.7	91.3 ± 2.1
TQA	85.4 ± 1.5	83.6 ± 1.7	91.3 ± 2.1	83.8 ± 7.3
TQA*	86.5 ± 1.5	84.2 ± 1.9	93.8 ± 1.4	88.5 ± 3.9
MPS (Quimb)				
Fixed Angle*	-	59.1 ± 3.3	97.8	96.1
Fixed Angle†	-	52.4 ± 3.3	97.4	90.9
Fourier*	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	88.8 ± 8.8
Interp.*	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	90.9 ± 7.3
Linear Ramp	95.5 ± 1.3	68.9 ± 3.6	94.1 ± 1.1	91.2 ± 1.9
Linear Ramp*	97.0 ± 1.1	72.1 ± 1.9	95.4 ± 1.0	92.9 ± 3.0
Recursive TS	-	-	-	-
TQA	83.7 ± 7.3	57.4 ± 1.1	90.5 ± 1.8	81.6 ± 12.0
TQA*	90.0 ± 2.3	57.8 ± 1.2	93.3 ± 1.0	90.1 ± 1.5
PP				
Fixed Angle*	-	93.7 ± 0.4	91.5	91.2
Fixed Angle†	-	90.9 ± 1.5	87.2	90.7
Fourier*	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	90.5 ± 2.3
Interp.*	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3
Linear Ramp	94.4 ± 1.0	92.2 ± 0.6	88.6 ± 1.8	90.8 ± 0.8
Linear Ramp*	95.6 ± 1.0	93.3 ± 0.5	90.7 ± 2.2	91.4 ± 0.8
Recursive TS	-	-	-	-
TQA	83.6 ± 0.9	89.1 ± 0.7	77.1 ± 7.9	86.9 ± 3.0
TQA*	92.9 ± 1.4	91.0 ± 0.6	85.6 ± 2.9	90.4 ± 0.9
SV				
Fixed Angle*	96.9 ± 1.4	-	98.4 ± 0.8	98.3 ± 1.1
Fixed Angle†	91.7 ± 3.5	-	97.0 ± 1.3	97.7 ± 1.0
Fourier*	78.3 ± 12.6	73.8 ± 10.4	78.9 ± 11.5	83.4 ± 13.8
Interp.*	96.2 ± 1.1	94.5 ± 1.1	95.5 ± 4.1	93.6 ± 6.1
Linear Ramp	93.7 ± 1.2	93.8 ± 0.5	92.8 ± 1.5	94.3 ± 1.5
Recursive TS	94.9 ± 1.9	92.5 ± 2.4	93.5 ± 5.1	93.6 ± 3.2
TQA	91.0 ± 3.6	88.9 ± 3.4	83.8 ± 9.2	90.8 ± 4.9
TQA*	94.9 ± 1.3	92.5 ± 1.6	93.6 ± 3.3	96.1 ± 1.7

Table 26: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	-	-	97.8	90.2	-	-	1 [100]	1 [100]
Fixed Angle†	-	-	97.5	85.1	-	-	1 [100]	1 [100]
Fourier*	78.1 ± 5.3	85.8 ± 4.0	79.4 ± 8.3	84.0 ± 9.3	20 [40–50]	20 [39–144]	49 [40–100]	17 [40–100]
Interp.*	94.3 ± 3.6	90.6 ± 4.8	94.7 ± 2.5	88.7 ± 7.1	20 [40–50]	20 [39–144]	49 [40–100]	17 [40–100]
Linear Ramp	92.8 ± 1.5	84.0 ± 2.7	94.6 ± 1.0	87.8 ± 2.7	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	95.4 ± 1.8	84.2 ± 2.0	96.0 ± 0.7	91.3 ± 2.1	10 [50]	10 [144]	10 [100]	7 [100]
TQA	85.4 ± 1.5	83.6 ± 1.7	91.3 ± 2.1	83.8 ± 7.3	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	86.5 ± 1.5	84.2 ± 1.9	93.8 ± 1.4	88.5 ± 3.9	10 [50]	10 [144]	10 [100]	7 [100]
MPS (Quimb)								
Fixed Angle*	-	59.1 ± 3.3	97.8	96.1	-	10 [144]	1 [100]	1 [100]
Fixed Angle†	-	52.4 ± 3.3	97.4	90.9	-	10 [144]	1 [100]	1 [100]
Fourier*	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	88.8 ± 8.8	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Interp.*	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	90.9 ± 7.3	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp	95.5 ± 1.3	68.9 ± 3.6	94.1 ± 1.1	91.2 ± 1.9	10 [50]	10 [144]	10 [100]	7 [100]
Linear Ramp*	97.0 ± 1.1	72.1 ± 1.9	95.4 ± 1.0	92.9 ± 3.0	10 [50]	10 [144]	10 [100]	7 [100]
Recursive TS	-	-	-	-	-	-	-	-
TQA	83.7 ± 7.3	57.4 ± 1.1	90.5 ± 1.8	81.6 ± 12.0	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	90.0 ± 2.3	57.8 ± 1.2	93.3 ± 1.0	90.1 ± 1.5	10 [50]	10 [144]	10 [100]	7 [100]
PP								
Fixed Angle*	-	93.7 ± 0.4	91.5	91.2	-	10 [144]	1 [100]	1 [100]
Fixed Angle†	-	90.9 ± 1.5	87.2	90.7	-	10 [144]	1 [100]	1 [100]
Fourier*	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	90.5 ± 2.3	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Interp.*	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3	20 [40–50]	31 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp	94.4 ± 1.0	92.2 ± 0.6	88.6 ± 1.8	90.8 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Linear Ramp*	95.6 ± 1.0	93.3 ± 0.5	90.7 ± 2.2	91.4 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	-	-	-	-	-	-	-	-
TQA	83.6 ± 0.9	89.1 ± 0.7	77.1 ± 7.9	86.9 ± 3.0	10 [50]	10 [144]	10 [100]	7 [100]
TQA*	92.9 ± 1.4	91.0 ± 0.6	85.6 ± 2.9	90.4 ± 0.9	10 [50]	10 [144]	10 [100]	7 [100]
SV								
Fixed Angle*	96.9 ± 1.4	-	98.4 ± 0.8	98.3 ± 1.1	17 [11–20]	-	19 [10–20]	50 [10–20]
Fixed Angle†	91.7 ± 3.5	-	97.0 ± 1.3	97.7 ± 1.0	17 [11–20]	-	19 [10–20]	50 [10–20]
Fourier*	78.3 ± 12.6	73.8 ± 10.4	78.9 ± 11.5	83.4 ± 13.8	90 [10–20]	20 [12–21]	776 [10–20]	364 [10–20]
Interp.*	96.2 ± 1.1	94.5 ± 1.1	95.5 ± 4.1	93.6 ± 6.1	91 [10–20]	20 [12–21]	777 [10–20]	371 [10–20]
Linear Ramp	93.7 ± 1.2	93.8 ± 0.5	92.8 ± 1.5	94.3 ± 1.5	10 [20]	10 [21]	10 [20]	7 [20]
Recursive TS	94.9 ± 1.9	92.5 ± 2.4	93.5 ± 5.1	93.6 ± 3.2	89 [10–20]	20 [12–21]	281 [10–20]	352 [10–20]
TQA	91.0 ± 3.6	88.9 ± 3.4	83.8 ± 9.2	90.8 ± 4.9	92 [10–20]	20 [12–21]	676 [10–20]	356 [10–20]
TQA*	94.9 ± 1.3	92.5 ± 1.6	93.6 ± 3.3	96.1 ± 1.7	90 [10–20]	20 [12–21]	671 [10–20]	141 [10–20]

Table 27: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

10 Tables for depth $P = 10$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	20 [39–144]	11 [100]	11 [40–100]
Fixed Angle†	-	20 [39–144]	11 [100]	11 [40–100]
Fourier*	20 [40–50]	20 [39–144]	49 [40–100]	11 [40–100]
Interp.*	20 [40–50]	20 [39–144]	49 [40–100]	11 [40–100]
Linear Ramp	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	1 [100]
TQA	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
MPS (Quimb)				
Fixed Angle*	-	-	1 [100]	12 [40–100]
Fixed Angle†	-	-	1 [100]	12 [40–100]
Fourier*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Interp.*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Linear Ramp	20 [40–50]	20 [39–144]	10 [100]	1 [100]
Linear Ramp*	10 [50]	10 [144]	10 [100]	1 [100]
Recursive TS	10 [50]	10 [144]	10 [100]	1 [100]
TQA	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
TQA*	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
PP				
Fixed Angle*	-	30 [39–144]	10 [100]	12 [40–100]
Fixed Angle†	-	30 [39–144]	10 [100]	12 [40–100]
Fourier*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Interp.*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp*	10 [50]	20 [105–144]	20 [40–100]	8 [100]
Recursive TS	10 [50]	10 [144]	10 [100]	7 [100]
TQA	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
SV				
Fixed Angle*	17 [11–20]	-	16 [10–20]	30 [10–20]
Fixed Angle†	17 [11–20]	-	16 [10–20]	30 [10–20]
Fourier*	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]
Interp.*	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]
Linear Ramp	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]
Recursive TS	90 [10–20]	20 [12–21]	180 [10–20]	147 [10–20]
TQA	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]
TQA*	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]

Table 28: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	-	89.7 ± 3.9	96.2 ± 1.0	98.5 ± 1.4
Fixed Angle [†]	-	88.3 ± 3.6	93.8 ± 2.1	97.8 ± 3.3
Fourier [*]	81.2 ± 6.1	85.1 ± 4.8	80.6 ± 8.9	90.1 ± 5.9
Interp. [*]	94.2 ± 4.0	91.3 ± 4.6	95.4 ± 2.4	91.1 ± 3.3
Linear Ramp	94.3 ± 1.9	92.1 ± 1.8	91.4 ± 4.8	95.4 ± 2.6
Linear Ramp [*]	96.2 ± 0.8	84.6 ± 1.3	96.2 ± 0.7	91.0
TQA	86.5 ± 2.1	87.1 ± 2.5	85.4 ± 4.4	92.7 ± 4.6
TQA [*]	89.4 ± 2.0	87.9 ± 2.9	89.8 ± 3.0	95.5 ± 3.5
MPS (Quimb)				
Fixed Angle [*]	-	-	98.1	98.4 ± 1.2
Fixed Angle [†]	-	-	97.7	97.7 ± 2.2
Fourier [*]	79.6 ± 6.1	77.4 ± 10.9	88.3 ± 5.6	91.8 ± 9.3
Interp. [*]	97.4 ± 1.5	86.6 ± 10.1	97.1 ± 0.6	92.9 ± 5.3
Linear Ramp	96.4 ± 0.9	82.2 ± 11.8	94.6 ± 1.1	88.7
Linear Ramp [*]	97.4 ± 0.7	69.5 ± 2.0	95.9 ± 1.0	92.7
Recursive TS	97.3 ± 1.1	56.5 ± 1.6	93.8 ± 1.4	88.4
TQA	87.7 ± 1.8	79.6 ± 9.2	90.5 ± 2.6	92.8 ± 5.2
TQA [*]	91.8 ± 2.9	80.8 ± 9.3	92.9 ± 1.8	96.1 ± 2.2
PP				
Fixed Angle [*]	-	94.2 ± 0.6	90.7 ± 1.9	93.5 ± 1.6
Fixed Angle [†]	-	91.5 ± 1.3	86.7 ± 1.7	92.9 ± 1.6
Fourier [*]	91.5 ± 1.9	91.3 ± 3.1	86.0 ± 2.2	90.7 ± 2.4
Interp. [*]	94.4 ± 1.9	95.0 ± 0.7	89.6 ± 1.7	90.4 ± 2.5
Linear Ramp	94.7 ± 1.2	93.0 ± 0.7	87.9 ± 1.7	91.9 ± 1.4
Linear Ramp [*]	95.7 ± 1.1	93.6 ± 0.5	91.0 ± 2.3	91.7 ± 0.8
Recursive TS	90.6 ± 1.1	89.8 ± 1.1	87.4 ± 2.0	87.4 ± 1.8
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	89.4 ± 3.0
TQA [*]	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	91.6 ± 1.6
SV				
Fixed Angle [*]	97.4 ± 1.4	-	97.5 ± 1.3	98.4 ± 0.9
Fixed Angle [†]	92.0 ± 3.6	-	96.3 ± 1.6	97.8 ± 0.9
Fourier [*]	77.9 ± 13.1	74.2 ± 12.3	76.8 ± 11.3	82.2 ± 14.3
Interp. [*]	96.5 ± 1.1	94.7 ± 1.0	95.4 ± 1.9	93.4 ± 6.4
Linear Ramp	95.0 ± 1.4	94.4 ± 1.0	93.5 ± 2.0	96.2 ± 1.8
Recursive TS	95.1 ± 2.0	92.8 ± 2.4	92.5 ± 2.2	94.1 ± 3.2
TQA	91.6 ± 3.6	89.8 ± 3.3	84.1 ± 8.2	91.3 ± 4.8
TQA [*]	95.4 ± 1.3	92.8 ± 1.9	93.2 ± 2.5	96.7 ± 1.5

Table 29: **Avg. Approx. Ratio ± standard deviation in percentage for MAXCUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	-	89.7 ± 3.9	96.2 ± 1.0	98.5 ± 1.4	-	20 [39–144]	11 [100]	11 [40–100]
Fixed Angle†	-	88.3 ± 3.6	93.8 ± 2.1	97.8 ± 3.3	-	20 [39–144]	11 [100]	11 [40–100]
Fourier*	81.2 ± 6.1	85.1 ± 4.8	80.6 ± 8.9	90.1 ± 5.9	20 [40–50]	20 [39–144]	49 [40–100]	11 [40–100]
Interp.*	94.2 ± 4.0	91.3 ± 4.6	95.4 ± 2.4	91.1 ± 3.3	20 [40–50]	20 [39–144]	49 [40–100]	11 [40–100]
Linear Ramp	94.3 ± 1.9	92.1 ± 1.8	91.4 ± 4.8	95.4 ± 2.6	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
Linear Ramp*	96.2 ± 0.8	84.6 ± 1.3	96.2 ± 0.7	91.0	10 [50]	10 [144]	10 [100]	1 [40–100]
TQA	86.5 ± 2.1	87.1 ± 2.5	85.4 ± 4.4	92.7 ± 4.6	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
TQA*	89.4 ± 2.0	87.9 ± 2.9	89.8 ± 3.0	95.5 ± 3.5	20 [40–50]	20 [39–144]	50 [40–100]	11 [40–100]
MPS (Quimb)								
Fixed Angle*	-	-	98.1	98.4 ± 1.2	-	-	1 [100]	12 [40–100]
Fixed Angle†	-	-	97.7	97.7 ± 2.2	-	-	1 [100]	12 [40–100]
Fourier*	79.6 ± 6.1	77.4 ± 10.9	88.3 ± 5.6	91.8 ± 9.3	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Interp.*	97.4 ± 1.5	86.6 ± 10.1	97.1 ± 0.6	92.9 ± 5.3	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
Linear Ramp	96.4 ± 0.9	82.2 ± 11.8	94.6 ± 1.1	88.7	20 [40–50]	20 [39–144]	10 [100]	1 [40–100]
Linear Ramp*	97.4 ± 0.7	69.5 ± 2.0	95.9 ± 1.0	92.7	10 [50]	10 [144]	10 [100]	1 [40–100]
Recursive TS	97.3 ± 1.1	56.5 ± 1.6	93.8 ± 1.4	88.4	10 [50]	10 [144]	10 [100]	1 [40–100]
TQA	87.7 ± 1.8	79.6 ± 9.2	90.5 ± 2.6	92.8 ± 5.2	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.9 ± 1.8	96.1 ± 2.2	20 [40–50]	31 [39–144]	10 [100]	12 [40–100]
PP								
Fixed Angle*	-	94.2 ± 0.6	90.7 ± 1.9	93.5 ± 1.6	-	30 [39–144]	10 [100]	12 [40–100]
Fixed Angle†	-	91.5 ± 1.3	86.7 ± 1.7	92.9 ± 1.6	-	30 [39–144]	10 [100]	12 [40–100]
Fourier*	91.5 ± 1.9	91.3 ± 3.1	86.0 ± 2.2	90.7 ± 2.4	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Interp.*	94.4 ± 1.9	95.0 ± 0.7	89.6 ± 1.7	90.4 ± 2.5	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp	94.7 ± 1.2	93.0 ± 0.7	87.9 ± 1.7	91.9 ± 1.4	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
Linear Ramp*	95.7 ± 1.1	93.6 ± 0.5	91.0 ± 2.3	91.7 ± 0.8	10 [50]	20 [105–144]	20 [40–100]	8 [40–100]
Recursive TS	90.6 ± 1.1	89.8 ± 1.1	87.4 ± 2.0	87.4 ± 1.8	10 [50]	10 [144]	10 [100]	7 [40–100]
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	89.4 ± 3.0	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	91.6 ± 1.6	20 [40–50]	30 [39–144]	111 [40–100]	18 [40–100]
SV								
Fixed Angle*	97.4 ± 1.4	-	97.5 ± 1.3	98.4 ± 0.9	17 [11–20]	-	16 [10–20]	30 [10–20]
Fixed Angle†	92.0 ± 3.6	-	96.3 ± 1.6	97.8 ± 0.9	17 [11–20]	-	16 [10–20]	30 [10–20]
Fourier*	77.9 ± 13.1	74.2 ± 12.3	76.8 ± 11.3	82.2 ± 14.3	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]
Interp.*	96.5 ± 1.1	94.7 ± 1.0	95.4 ± 1.9	93.4 ± 6.4	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]
Linear Ramp	95.0 ± 1.4	94.4 ± 1.0	93.5 ± 2.0	96.2 ± 1.8	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]
Recursive TS	95.1 ± 2.0	92.8 ± 2.4	92.5 ± 2.2	94.1 ± 3.2	90 [10–20]	20 [12–21]	180 [10–20]	147 [10–20]
TQA	91.6 ± 3.6	89.8 ± 3.3	84.1 ± 8.2	91.3 ± 4.8	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]
TQA*	95.4 ± 1.3	92.8 ± 1.9	93.2 ± 2.5	96.7 ± 1.5	90 [10–20]	20 [12–21]	180 [10–20]	153 [10–20]

Table 30: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for MAX-CUT for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

Summary Tables of MIS Data

11 Tables for depth $P = 1$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	10 [70]	10 [144]	10 [100]	10 [100]
TQA	-	-	-	-
TQA*	-	-	-	-
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	10 [70]	10 [144]	10 [100]	10 [100]
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	-	-	-	10 [20]
Fixed Angle†	-	-	-	10 [20]
Fourier*	8 [20]	-	8 [20]	10 [20]
Interp.*	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-
Recursive TS	7 [20]	-	8 [20]	10 [20]
TQA	-	-	-	10 [20]
TQA*	-	-	-	10 [20]

Table 31: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9229.9 ± 7.4
Interp. [*]	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9229.9 ± 7.4
TQA	-	-	-	-
TQA [*]	-	-	-	-
MPS (Quimb)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	-	-	-
Interp. [*]	-	-	-	-
TQA	-	-	-	-
TQA [*]	-	-	-	-
PP				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9229.9 ± 7.4
Interp. [*]	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9229.9 ± 7.4
TQA	-	-	-	-
TQA [*]	-	-	-	-
SV				
Fixed Angle [*]	-	-	-	1857.0 ± 11.0
Fixed Angle [†]	-	-	-	1217.6 ± 2.6
Fourier [*]	2736.1 ± 507.1	-	2852.0 ± 0.0	1857.0 ± 11.0
Interp. [*]	2720.6 ± 545.7	-	2852.0 ± 0.0	1857.0 ± 11.0
Linear Ramp	-	-	-	-
Recursive TS	2720.6 ± 545.7	-	2852.0 ± 0.0	1857.0 ± 11.0
TQA	-	-	-	1789.5 ± 10.7
TQA [*]	-	-	-	1857.0 ± 11.0

Table 32: **Avg. Energy ± standard deviation for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-	-
Fourier*	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9229.9 ± 7.4	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9229.9 ± 7.4	10 [70]	10 [144]	10 [100]	10 [100]
TQA	-	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-	-
MPS (Quimb)								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-	-
Fourier*	-	-	-	-	-	-	-	-
Interp.*	-	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-	-
PP								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-	-
Fourier*	56682.2 ± 2535.8	7484.9 ± 0.0	15317.6 ± 0.0	9229.9 ± 7.4	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	56598.1 ± 2542.4	7484.9 ± 0.0	15317.6 ± 0.0	9229.9 ± 7.4	10 [70]	10 [144]	10 [100]	10 [100]
TQA	-	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-	-
SV								
Fixed Angle*	-	-	-	1857.0 ± 11.0	-	-	-	10 [20]
Fixed Angle†	-	-	-	1217.6 ± 2.6	-	-	-	10 [20]
Fourier*	2736.1 ± 507.1	-	2852.0 ± 0.0	1857.0 ± 11.0	8 [20]	-	8 [20]	10 [20]
Interp.*	2720.6 ± 545.7	-	2852.0 ± 0.0	1857.0 ± 11.0	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-	-	-	-	-
Recursive TS	2720.6 ± 545.7	-	2852.0 ± 0.0	1857.0 ± 11.0	7 [20]	-	8 [20]	10 [20]
TQA	-	-	-	1789.5 ± 10.7	-	-	-	10 [20]
TQA*	-	-	-	1857.0 ± 11.0	-	-	-	10 [20]

Table 33: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

12 Tables for depth $P = 2$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	10 [70]	10 [144]	10 [100]	10 [100]
Fixed Angle†	10 [70]	10 [144]	10 [100]	10 [100]
Fourier*	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	10 [70]	10 [144]	10 [100]	10 [100]
TQA	10 [70]	10 [144]	10 [100]	10 [100]
TQA*	10 [70]	10 [144]	10 [100]	10 [100]
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	10 [144]	-	-
TQA*	-	10 [144]	-	-
PP				
Fixed Angle*	10 [70]	10 [144]	10 [100]	10 [100]
Fixed Angle†	10 [70]	10 [144]	10 [100]	10 [100]
Fourier*	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	10 [70]	10 [144]	10 [100]	10 [100]
TQA	10 [70]	10 [144]	10 [100]	10 [100]
TQA*	10 [70]	10 [144]	10 [100]	10 [100]
SV				
Fixed Angle*	-	-	-	10 [20]
Fixed Angle†	-	-	-	10 [20]
Fourier*	8 [20]	-	8 [20]	10 [20]
Interp.*	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	10 [20]
TQA*	-	-	-	10 [20]

Table 34: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	11603.1 $\pm$ 5689.7	15086.0 $\pm$ 21.2	20451.5 $\pm$ 221.4	14594.1 $\pm$ 163.9
Fixed Angle [†]	3358.9 $\pm$ 3731.9	8967.4 $\pm$ 17.4	17687.2 $\pm$ 21.1	6117.7 $\pm$ 67.3
Fourier*	61936.6 $\pm$ 2524.1	15087.0 $\pm$ 19.3	23286.0 $\pm$ 577.7	16822.9 $\pm$ 133.1
Interp.*	53447.9 $\pm$ 6357.5	15047.8 $\pm$ 53.3	19285.1 $\pm$ 523.1	16950.8 $\pm$ 64.1
TQA	26859.1 $\pm$ 3717.5	11388.1 $\pm$ 8.8	21309.1 $\pm$ 17.0	15015.9 $\pm$ 134.0
TQA*	59220.7 $\pm$ 3629.8	15043.9 $\pm$ 77.7	23709.2 $\pm$ 162.9	16962.7 $\pm$ 77.7
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	13385.3 $\pm$ 0.0	-	-
TQA*	-	15348.6 $\pm$ 272.1	-	-
PP				
Fixed Angle*	1462.7 $\pm$ 1983.1	14668.7 $\pm$ 0.0	19901.6 $\pm$ 0.0	12235.0 $\pm$ 22.2
Fixed Angle [†]	-159.9 $\pm$ 625.3	8262.0 $\pm$ 0.0	11104.2 $\pm$ 0.0	6236.1 $\pm$ 12.9
Fourier*	66023.6 $\pm$ 2531.7	14667.9 $\pm$ 0.0	23962.1 $\pm$ 0.0	16649.5 $\pm$ 7.4
Interp.*	53045.1 $\pm$ 6171.8	14668.8 $\pm$ 0.0	20205.4 $\pm$ 0.0	16648.6 $\pm$ 7.9
TQA	23622.4 $\pm$ 2109.4	11591.9 $\pm$ 0.0	21688.0 $\pm$ 0.0	14559.1 $\pm$ 5.9
TQA*	64848.8 $\pm$ 5913.4	14668.5 $\pm$ 0.0	23961.1 $\pm$ 0.0	16645.6 $\pm$ 8.6
SV				
Fixed Angle*	-	-	-	2494.3 $\pm$ 22.7
Fixed Angle [†]	-	-	-	1289.9 $\pm$ 28.1
Fourier*	3847.9 $\pm$ 652.5	-	4329.1 $\pm$ 0.0	3294.5 $\pm$ 10.2
Interp.*	3735.3 $\pm$ 636.3	-	2465.7 $\pm$ 0.0	3294.0 $\pm$ 12.9
Linear Ramp	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	2912.4 $\pm$ 6.2
TQA*	-	-	-	3315.8 $\pm$ 8.7

Table 35: **Avg. Energy $\pm$ standard deviation for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fixed Angle*	11603.1 $\pm$ 5689.7	15086.0 $\pm$ 21.2	20451.5 $\pm$ 221.4	14594.1 $\pm$ 163.9	10 [70]	10 [144]	10 [100]
Fixed Angle [†]	3358.9 $\pm$ 3731.9	8967.4 $\pm$ 17.4	17687.2 $\pm$ 21.1	6117.7 $\pm$ 67.3	10 [70]	10 [144]	10 [100]
Fourier*	61936.6 $\pm$ 2524.1	15087.0 $\pm$ 19.3	23286.0 $\pm$ 577.7	16822.9 $\pm$ 133.1	10 [70]	10 [144]	10 [100]
Interp.*	53447.9 $\pm$ 6357.5	15047.8 $\pm$ 53.3	19285.1 $\pm$ 523.1	16950.8 $\pm$ 64.1	10 [70]	10 [144]	10 [100]
TQA	26859.1 $\pm$ 3717.5	11388.1 $\pm$ 8.8	21309.1 $\pm$ 17.0	15015.9 $\pm$ 134.0	10 [70]	10 [144]	10 [100]
TQA*	59220.7 $\pm$ 3629.8	15043.9 $\pm$ 77.7	23709.2 $\pm$ 162.9	16962.7 $\pm$ 77.7	10 [70]	10 [144]	10 [100]
MPS (Quimb)							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-
Fourier*	-	-	-	-	-	-	-
Interp.*	-	-	-	-	-	-	-
TQA	-	13385.3 $\pm$ 0.0	-	-	-	10 [144]	-
TQA*	-	15348.6 $\pm$ 272.1	-	-	-	10 [144]	-
PP							
Fixed Angle*	1462.7 $\pm$ 1983.1	14668.7 $\pm$ 0.0	19901.6 $\pm$ 0.0	12235.0 $\pm$ 22.2	10 [70]	10 [144]	10 [100]
Fixed Angle [†]	-159.9 $\pm$ 625.3	8262.0 $\pm$ 0.0	11104.2 $\pm$ 0.0	6236.1 $\pm$ 12.9	10 [70]	10 [144]	10 [100]
Fourier*	66023.6 $\pm$ 2531.7	14667.9 $\pm$ 0.0	23962.1 $\pm$ 0.0	16649.5 $\pm$ 7.4	10 [70]	10 [144]	10 [100]
Interp.*	53045.1 $\pm$ 6171.8	14668.8 $\pm$ 0.0	20205.4 $\pm$ 0.0	16648.6 $\pm$ 7.9	10 [70]	10 [144]	10 [100]
TQA	23622.4 $\pm$ 2109.4	11591.9 $\pm$ 0.0	21688.0 $\pm$ 0.0	14559.1 $\pm$ 5.9	10 [70]	10 [144]	10 [100]
TQA*	64848.8 $\pm$ 5913.4	14668.5 $\pm$ 0.0	23961.1 $\pm$ 0.0	16645.6 $\pm$ 8.6	10 [70]	10 [144]	10 [100]
SV							
Fixed Angle*	-	-	-	2494.3 $\pm$ 22.7	-	-	-
Fixed Angle [†]	-	-	-	1289.9 $\pm$ 28.1	-	-	-
Fourier*	3847.9 $\pm$ 652.5	-	4329.1 $\pm$ 0.0	3294.5 $\pm$ 10.2	8 [20]	-	8 [20]
Interp.*	3735.3 $\pm$ 636.3	-	2465.7 $\pm$ 0.0	3294.0 $\pm$ 12.9	7 [20]	-	8 [20]
Linear Ramp	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	-	-	-	2912.4 $\pm$ 6.2	-	-	-
TQA*	-	-	-	3315.8 $\pm$ 8.7	-	-	-

Table 36: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

13 Tables for depth $P = 3$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	10 [70]	10 [144]	10 [100]	10 [100]
TQA	-	-	-	-
TQA*	-	-	-	-
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	10 [70]	10 [144]	10 [100]	10 [100]
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	-	-	-	10 [20]
Fixed Angle†	-	-	-	10 [20]
Fourier*	8 [20]	-	8 [20]	10 [20]
Interp.*	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-
Recursive TS	7 [20]	-	8 [20]	-
TQA	-	-	-	10 [20]
TQA*	-	-	-	10 [20]

Table 37: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	63673.4 ± 2329.8	16820.8 ± 51.2	23195.5 ± 513.9	17145.6 ± 166.6
Interp.*	30291.4 ± 8828.4	15646.5 ± 437.4	20601.2 ± 388.8	17142.0 ± 124.0
TQA	-	-	-	-
TQA*	-	-	-	-
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	70899.1 ± 2921.4	16638.6 ± 0.0	25141.6 ± 0.0	17514.9 ± 4.0
Interp.*	15399.2 ± 5117.8	16615.9 ± 0.0	23006.9 ± 0.0	17490.7 ± 13.7
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	-	-	-	3310.9 ± 10.3
Fixed Angle†	-	-	-	929.2 ± 17.4
Fourier*	4196.9 ± 714.4	-	4443.3 ± 0.0	3335.3 ± 47.3
Interp.*	3838.4 ± 626.5	-	3382.5 ± 0.0	3232.8 ± 18.9
Linear Ramp	-	-	-	-
Recursive TS	4259.3 ± 598.9	-	4378.7 ± 0.0	-
TQA	-	-	-	3260.1 ± 9.3
TQA*	-	-	-	3479.4 ± 5.4

Table 38: **Avg. Energy ± standard deviation for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-
Fourier*	63673.4 ± 2329.8	16820.8 ± 51.2	23195.5 ± 513.9	17145.6 ± 166.6	10 [70]	10 [144]	10 [100]
Interp.*	30291.4 ± 8828.4	15646.5 ± 437.4	20601.2 ± 388.8	17142.0 ± 124.0	10 [70]	10 [144]	10 [100]
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
MPS (Quimb)							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-
Fourier*	-	-	-	-	-	-	-
Interp.*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
PP							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-
Fourier*	70899.1 ± 2921.4	16638.6 ± 0.0	25141.6 ± 0.0	17514.9 ± 4.0	10 [70]	10 [144]	10 [100]
Interp.*	15399.2 ± 5117.8	16615.9 ± 0.0	23006.9 ± 0.0	17490.7 ± 13.7	10 [70]	10 [144]	10 [100]
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
SV							
Fixed Angle*	-	-	-	3310.9 ± 10.3	-	-	-
Fixed Angle†	-	-	-	929.2 ± 17.4	-	-	-
Fourier*	4196.9 ± 714.4	-	4443.3 ± 0.0	3335.3 ± 47.3	8 [20]	-	8 [20]
Interp.*	3838.4 ± 626.5	-	3382.5 ± 0.0	3232.8 ± 18.9	7 [20]	-	8 [20]
Linear Ramp	-	-	-	-	-	-	-
Recursive TS	4259.3 ± 598.9	-	4378.7 ± 0.0	-	7 [20]	-	8 [20]
TQA	-	-	-	3260.1 ± 9.3	-	-	-
TQA*	-	-	-	3479.4 ± 5.4	-	-	-

Table 39: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

14 Tables for depth $P = 4$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	10 [70]	10 [144]	10 [100]	10 [100]
Fixed Angle [†]	10 [70]	10 [144]	10 [100]	10 [100]
Fourier [*]	10 [70]	10 [144]	10 [100]	10 [100]
Interp. [*]	10 [70]	10 [144]	10 [100]	10 [100]
TQA	10 [70]	10 [144]	10 [100]	10 [100]
TQA [*]	10 [70]	10 [144]	10 [100]	10 [100]
MPS (Quimb)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	-	-	-
Interp. [*]	-	-	-	-
TQA	-	10 [144]	-	-
TQA [*]	-	10 [144]	-	-
PP				
Fixed Angle [*]	10 [70]	10 [144]	10 [100]	10 [100]
Fixed Angle [†]	10 [70]	10 [144]	10 [100]	10 [100]
Fourier [*]	10 [70]	10 [144]	10 [100]	10 [100]
Interp. [*]	10 [70]	10 [144]	10 [100]	10 [100]
TQA	10 [70]	10 [144]	10 [100]	10 [100]
TQA [*]	10 [70]	10 [144]	10 [100]	10 [100]
SV				
Fixed Angle [*]	-	-	-	10 [20]
Fixed Angle [†]	-	-	-	10 [20]
Fourier [*]	8 [20]	-	8 [20]	10 [20]
Interp. [*]	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-
Recursive TS	7 [20]	-	8 [20]	-
TQA	-	-	-	10 [20]
TQA [*]	-	-	-	10 [20]

Table 40: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	11995.1 ± 3162.9	16523.3 ± 100.5	22494.0 ± 518.0	17239.6 ± 49.8
Fixed Angle†	-12244.9 ± 3869.0	13278.6 ± 16.9	8579.6 ± 19.7	14350.8 ± 279.9
Fourier*	65073.4 ± 2065.2	17025.1 ± 65.9	23889.9 ± 331.7	17227.5 ± 109.2
Interp.*	30289.6 ± 18260.2	16319.8 ± 349.4	22168.3 ± 982.7	17355.5 ± 48.4
TQA	45337.4 ± 4065.1	15942.9 ± 6.2	21463.2 ± 18.7	16587.7 ± 14.7
TQA*	52894.8 ± 6299.2	16825.6 ± 75.2	23803.7 ± 314.5	17309.4 ± 25.9
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	15660.4 ± 8.7	-	-
TQA*	-	16122.0 ± 113.1	-	-
PP				
Fixed Angle*	6774.6 ± 6735.8	15702.5 ± 0.0	20203.6 ± 0.0	16592.8 ± 16.9
Fixed Angle†	-2308.7 ± 383.9	10682.9 ± 0.0	5827.3 ± 0.0	6629.2 ± 24.3
Fourier*	73002.8 ± 3089.2	17057.3 ± 0.0	25757.2 ± 0.0	17619.5 ± 15.3
Interp.*	27879.7 ± 27264.2	16074.5 ± 0.0	23790.6 ± 0.0	17580.0 ± 17.3
TQA	18316.7 ± 5870.9	16094.1 ± 0.0	21509.8 ± 0.0	16273.7 ± 4.3
TQA*	57385.8 ± 12377.5	16902.2 ± 0.0	24749.9 ± 0.0	17512.7 ± 19.7
SV				
Fixed Angle*	-	-	-	3325.0 ± 11.3
Fixed Angle†	-	-	-	1332.7 ± 20.0
Fourier*	4450.6 ± 639.3	-	4595.6 ± 0.0	3466.2 ± 8.9
Interp.*	3961.1 ± 561.2	-	3803.1 ± 0.0	3444.1 ± 5.1
Linear Ramp	-	-	-	-
Recursive TS	4397.5 ± 619.9	-	4446.1 ± 0.0	-
TQA	-	-	-	3259.8 ± 3.1
TQA*	-	-	-	3479.2 ± 1.4

Table 41: **Avg. Energy $\pm$ standard deviation for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fixed Angle*	11995.1 $\pm$ 3162.9	16523.3 $\pm$ 100.5	22494.0 $\pm$ 518.0	17239.6 $\pm$ 49.8	10 [70]	10 [144]	10 [100]
Fixed Angle [†]	−12244.9 $\pm$ 3869.0	13278.6 $\pm$ 16.9	8579.6 $\pm$ 19.7	14350.8 $\pm$ 279.9	10 [70]	10 [144]	10 [100]
Fourier*	65073.4 $\pm$ 2065.2	17025.1 $\pm$ 65.9	23889.9 $\pm$ 331.7	17227.5 $\pm$ 109.2	10 [70]	10 [144]	10 [100]
Interp.*	30289.6 $\pm$ 18260.2	16319.8 $\pm$ 349.4	22168.3 $\pm$ 982.7	17355.5 $\pm$ 48.4	10 [70]	10 [144]	10 [100]
TQA	45337.4 $\pm$ 4065.1	15942.9 $\pm$ 6.2	21463.2 $\pm$ 18.7	16587.7 $\pm$ 14.7	10 [70]	10 [144]	10 [100]
TQA*	52894.8 $\pm$ 6299.2	16825.6 $\pm$ 75.2	23803.7 $\pm$ 314.5	17309.4 $\pm$ 25.9	10 [70]	10 [144]	10 [100]
MPS (Quimb)							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-
Fourier*	-	-	-	-	-	-	-
Interp.*	-	-	-	-	-	-	-
TQA	-	15660.4 $\pm$ 8.7	-	-	-	10 [144]	-
TQA*	-	16122.0 $\pm$ 113.1	-	-	-	10 [144]	-
PP							
Fixed Angle*	6774.6 $\pm$ 6735.8	15702.5 $\pm$ 0.0	20203.6 $\pm$ 0.0	16592.8 $\pm$ 16.9	10 [70]	10 [144]	10 [100]
Fixed Angle [†]	−2308.7 $\pm$ 383.9	10682.9 $\pm$ 0.0	5827.3 $\pm$ 0.0	6629.2 $\pm$ 24.3	10 [70]	10 [144]	10 [100]
Fourier*	73002.8 $\pm$ 3089.2	17057.3 $\pm$ 0.0	25757.2 $\pm$ 0.0	17619.5 $\pm$ 15.3	10 [70]	10 [144]	10 [100]
Interp.*	27879.7 $\pm$ 27264.2	16074.5 $\pm$ 0.0	23790.6 $\pm$ 0.0	17580.0 $\pm$ 17.3	10 [70]	10 [144]	10 [100]
TQA	18316.7 $\pm$ 5870.9	16094.1 $\pm$ 0.0	21509.8 $\pm$ 0.0	16273.7 $\pm$ 4.3	10 [70]	10 [144]	10 [100]
TQA*	57385.8 $\pm$ 12377.5	16902.2 $\pm$ 0.0	24749.9 $\pm$ 0.0	17512.7 $\pm$ 19.7	10 [70]	10 [144]	10 [100]
SV							
Fixed Angle*	-	-	-	3325.0 $\pm$ 11.3	-	-	-
Fixed Angle [†]	-	-	-	1332.7 $\pm$ 20.0	-	-	-
Fourier*	4450.6 $\pm$ 639.3	-	4595.6 $\pm$ 0.0	3466.2 $\pm$ 8.9	8 [20]	-	8 [20]
Interp.*	3961.1 $\pm$ 561.2	-	3803.1 $\pm$ 0.0	3444.1 $\pm$ 5.1	7 [20]	-	8 [20]
Linear Ramp	-	-	-	-	-	-	-
Recursive TS	4397.5 $\pm$ 619.9	-	4446.1 $\pm$ 0.0	-	7 [20]	-	8 [20]
TQA	-	-	-	3259.8 $\pm$ 3.1	-	-	-
TQA*	-	-	-	3479.2 $\pm$ 1.4	-	-	-

Table 42: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

15 Tables for depth $P = 5$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	10 [70]	10 [144]	10 [100]	10 [100]
TQA	-	-	-	-
TQA*	-	-	-	-
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	10 [70]	10 [144]	10 [100]	10 [100]
Interp.*	10 [70]	10 [144]	10 [100]	10 [100]
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	-	-	-	10 [20]
Fixed Angle†	-	-	-	10 [20]
Fourier*	8 [20]	-	8 [20]	10 [20]
Interp.*	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-
Recursive TS	7 [20]	-	8 [20]	-
TQA	-	-	-	10 [20]
TQA*	-	-	-	10 [20]

Table 43: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	65502.3 ± 2332.4	17150.3 ± 42.1	24101.5 ± 259.9	17355.1 ± 70.2
Interp.*	27765.9 ± 13747.4	16526.5 ± 257.7	22601.5 ± 712.6	17424.0 ± 38.1
TQA	-	-	-	-
TQA*	-	-	-	-
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	79758.6 ± 2424.4	17143.2 ± 0.0	26076.0 ± 0.0	17773.9 ± 20.7
Interp.*	35592.9 ± 25747.1	16105.4 ± 0.0	19595.6 ± 0.0	17756.7 ± 27.4
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	-	-	-	3184.5 ± 176.6
Fixed Angle†	-	-	-	1129.2 ± 9.2
Fourier*	4583.4 ± 615.3	-	4674.3 ± 0.0	3478.5 ± 5.5
Interp.*	3985.2 ± 509.3	-	4171.2 ± 0.0	3462.5 ± 3.6
Linear Ramp	-	-	-	-
Recursive TS	4459.5 ± 634.6	-	4461.3 ± 0.0	-
TQA	-	-	-	3297.7 ± 5.7
TQA*	-	-	-	3479.0 ± 1.9

Table 44: **Avg. Energy ± standard deviation for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-
Fourier*	65502.3 ± 2332.4	17150.3 ± 42.1	24101.5 ± 259.9	17355.1 ± 70.2	10 [70]	10 [144]	10 [100]
Interp.*	27765.9 ± 13747.4	16526.5 ± 257.7	22601.5 ± 712.6	17424.0 ± 38.1	10 [70]	10 [144]	10 [100]
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
MPS (Quimb)							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-
Fourier*	-	-	-	-	-	-	-
Interp.*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
PP							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-
Fourier*	79758.6 ± 2424.4	17143.2 ± 0.0	26076.0 ± 0.0	17773.9 ± 20.7	10 [70]	10 [144]	10 [100]
Interp.*	35592.9 ± 25747.1	16105.4 ± 0.0	19595.6 ± 0.0	17756.7 ± 27.4	10 [70]	10 [144]	10 [100]
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
SV							
Fixed Angle*	-	-	-	3184.5 ± 176.6	-	-	-
Fixed Angle†	-	-	-	1129.2 ± 9.2	-	-	-
Fourier*	4583.4 ± 615.3	-	4674.3 ± 0.0	3478.5 ± 5.5	8 [20]	-	8 [20]
Interp.*	3985.2 ± 509.3	-	4171.2 ± 0.0	3462.5 ± 3.6	7 [20]	-	8 [20]
Linear Ramp	-	-	-	-	-	-	-
Recursive TS	4459.5 ± 634.6	-	4461.3 ± 0.0	-	7 [20]	-	8 [20]
TQA	-	-	-	3297.7 ± 5.7	-	-	-
TQA*	-	-	-	3479.0 ± 1.9	-	-	-

Table 45: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

16 Tables for depth $P = 6$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	10 [70]	10 [144]	10 [100]	10 [100]
Fixed Angle [†]	10 [70]	10 [144]	10 [100]	10 [100]
Fourier [*]	10 [70]	10 [144]	10 [100]	10 [100]
Interp. [*]	10 [70]	10 [144]	10 [100]	10 [100]
TQA	10 [70]	10 [144]	10 [100]	10 [100]
TQA [*]	10 [70]	10 [144]	10 [100]	10 [100]
MPS (Quimb)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	-	-	-
Interp. [*]	-	-	-	-
TQA	-	10 [144]	-	-
TQA [*]	-	10 [144]	-	-
PP				
Fixed Angle [*]	10 [70]	10 [144]	10 [100]	10 [100]
Fixed Angle [†]	10 [70]	10 [144]	10 [100]	10 [100]
Fourier [*]	10 [70]	10 [144]	10 [100]	10 [100]
Interp. [*]	10 [70]	10 [144]	10 [100]	10 [100]
TQA	10 [70]	10 [144]	10 [100]	10 [100]
TQA [*]	10 [70]	10 [144]	10 [100]	10 [100]
SV				
Fixed Angle [*]	1 [20]	-	-	10 [20]
Fixed Angle [†]	1 [20]	-	-	10 [20]
Fourier [*]	8 [20]	-	8 [20]	10 [20]
Interp. [*]	7 [20]	-	8 [20]	10 [20]
Linear Ramp	10 [20]	10 [21]	10 [20]	10 [20]
Recursive TS	7 [20]	-	8 [20]	-
TQA	10 [20]	10 [21]	10 [20]	10 [20]
TQA [*]	10 [20]	10 [21]	10 [20]	10 [20]

Table 46: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	16048.7 $\pm$ 7754.3	17011.0 $\pm$ 37.3	22541.1 $\pm$ 385.0	17173.4 $\pm$ 50.6
Fixed Angle [†]	-18046.0 $\pm$ 3622.7	13274.4 $\pm$ 13.5	7168.5 $\pm$ 29.9	16194.0 $\pm$ 92.0
Fourier*	65882.6 $\pm$ 2605.2	17238.8 $\pm$ 41.9	24117.0 $\pm$ 532.0	17398.2 $\pm$ 62.9
Interp.*	41477.0 $\pm$ 10764.6	16407.2 $\pm$ 251.6	22849.9 $\pm$ 266.4	17427.2 $\pm$ 68.3
TQA	12617.7 $\pm$ 2917.7	16331.0 $\pm$ 4.1	21160.4 $\pm$ 21.2	16771.7 $\pm$ 14.6
TQA*	32061.9 $\pm$ 11764.8	17136.6 $\pm$ 34.3	24234.3 $\pm$ 142.4	17309.3 $\pm$ 40.8
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	16300.9 $\pm$ 0.7	-	-
TQA*	-	17190.4 $\pm$ 15.8	-	-
PP				
Fixed Angle*	13159.7 $\pm$ 13436.7	16922.4 $\pm$ 0.0	23810.2 $\pm$ 0.0	17248.7 $\pm$ 57.6
Fixed Angle [†]	-1659.8 $\pm$ 263.1	11040.6 $\pm$ 0.0	1593.1 $\pm$ 0.0	10487.2 $\pm$ 18.4
Fourier*	78103.2 $\pm$ 7976.1	17172.0 $\pm$ 0.0	25853.4 $\pm$ 0.0	17940.8 $\pm$ 30.8
Interp.*	55315.5 $\pm$ 32067.7	16369.2 $\pm$ 0.0	25263.7 $\pm$ 0.0	17786.4 $\pm$ 18.8
TQA	3285.8 $\pm$ 753.9	15990.3 $\pm$ 0.0	18897.2 $\pm$ 0.0	16844.9 $\pm$ 5.5
TQA*	19827.1 $\pm$ 3918.3	17075.8 $\pm$ 0.0	25357.7 $\pm$ 0.0	17627.0 $\pm$ 81.7
SV				
Fixed Angle*	3492.7	-	-	3390.5 $\pm$ 39.3
Fixed Angle [†]	611.2	-	-	2074.4 $\pm$ 28.2
Fourier*	4642.4 $\pm$ 600.2	-	4673.3 $\pm$ 0.0	3468.6 $\pm$ 13.6
Interp.*	3998.6 $\pm$ 506.1	-	3526.6 $\pm$ 0.0	3460.2 $\pm$ 3.6
Linear Ramp	4777.5 $\pm$ 779.0	2107.4 $\pm$ 0.0	4526.2 $\pm$ 0.0	3358.8 $\pm$ 6.6
Recursive TS	4498.8 $\pm$ 613.1	-	4461.7 $\pm$ 0.0	-
TQA	2644.2 $\pm$ 630.6	2126.2 $\pm$ 0.0	3539.2 $\pm$ 0.0	3304.4 $\pm$ 1.8
TQA*	4239.2 $\pm$ 561.7	2228.6 $\pm$ 0.0	4634.9 $\pm$ 0.0	3481.8 $\pm$ 3.1

Table 47: **Avg. Energy $\pm$ standard deviation for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances		
	ER	HH	LB	RR	ER	HH	LB
MPS (Aer)							
Fixed Angle*	16048.7 $\pm$ 7754.3	17011.0 $\pm$ 37.3	22541.1 $\pm$ 385.0	17173.4 $\pm$ 50.6	10 [70]	10 [144]	10 [100]
Fixed Angle [†]	−18046.0 $\pm$ 3622.7	13274.4 $\pm$ 13.5	7168.5 $\pm$ 29.9	16194.0 $\pm$ 92.0	10 [70]	10 [144]	10 [100]
Fourier*	65882.6 $\pm$ 2605.2	17238.8 $\pm$ 41.9	24117.0 $\pm$ 532.0	17398.2 $\pm$ 62.9	10 [70]	10 [144]	10 [100]
Interp.*	41477.0 $\pm$ 10764.6	16407.2 $\pm$ 251.6	22849.9 $\pm$ 266.4	17427.2 $\pm$ 68.3	10 [70]	10 [144]	10 [100]
TQA	12617.7 $\pm$ 2917.7	16331.0 $\pm$ 4.1	21160.4 $\pm$ 21.2	16771.7 $\pm$ 14.6	10 [70]	10 [144]	10 [100]
TQA*	32061.9 $\pm$ 11764.8	17136.6 $\pm$ 34.3	24234.3 $\pm$ 142.4	17309.3 $\pm$ 40.8	10 [70]	10 [144]	10 [100]
MPS (Quimb)							
Fixed Angle*	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-
Fourier*	-	-	-	-	-	-	-
Interp.*	-	-	-	-	-	-	-
TQA	-	16300.9 $\pm$ 0.7	-	-	-	10 [144]	-
TQA*	-	17190.4 $\pm$ 15.8	-	-	-	10 [144]	-
PP							
Fixed Angle*	13159.7 $\pm$ 13436.7	16922.4 $\pm$ 0.0	23810.2 $\pm$ 0.0	17248.7 $\pm$ 57.6	10 [70]	10 [144]	10 [100]
Fixed Angle [†]	−1659.8 $\pm$ 263.1	11040.6 $\pm$ 0.0	1593.1 $\pm$ 0.0	10487.2 $\pm$ 18.4	10 [70]	10 [144]	10 [100]
Fourier*	78103.2 $\pm$ 7976.1	17172.0 $\pm$ 0.0	25853.4 $\pm$ 0.0	17940.8 $\pm$ 30.8	10 [70]	10 [144]	10 [100]
Interp.*	55315.5 $\pm$ 32067.7	16369.2 $\pm$ 0.0	25263.7 $\pm$ 0.0	17786.4 $\pm$ 18.8	10 [70]	10 [144]	10 [100]
TQA	3285.8 $\pm$ 753.9	15990.3 $\pm$ 0.0	18897.2 $\pm$ 0.0	16844.9 $\pm$ 5.5	10 [70]	10 [144]	10 [100]
TQA*	19827.1 $\pm$ 3918.3	17075.8 $\pm$ 0.0	25357.7 $\pm$ 0.0	17627.0 $\pm$ 81.7	10 [70]	10 [144]	10 [100]
SV							
Fixed Angle*	3492.7	-	-	3390.5 $\pm$ 39.3	1 [20]	-	-
Fixed Angle [†]	611.2	-	-	2074.4 $\pm$ 28.2	1 [20]	-	-
Fourier*	4642.4 $\pm$ 600.2	-	4673.3 $\pm$ 0.0	3468.6 $\pm$ 13.6	8 [20]	-	8 [20]
Interp.*	3998.6 $\pm$ 506.1	-	3526.6 $\pm$ 0.0	3460.2 $\pm$ 3.6	7 [20]	-	8 [20]
Linear Ramp	4777.5 $\pm$ 779.0	2107.4 $\pm$ 0.0	4526.2 $\pm$ 0.0	3358.8 $\pm$ 6.6	10 [20]	10 [21]	10 [20]
Recursive TS	4498.8 $\pm$ 613.1	-	4461.7 $\pm$ 0.0	-	7 [20]	-	8 [20]
TQA	2644.2 $\pm$ 630.6	2126.2 $\pm$ 0.0	3539.2 $\pm$ 0.0	3304.4 $\pm$ 1.8	10 [20]	10 [21]	10 [20]
TQA*	4239.2 $\pm$ 561.7	2228.6 $\pm$ 0.0	4634.9 $\pm$ 0.0	3481.8 $\pm$ 3.1	10 [20]	10 [21]	10 [20]

Table 48: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

17 Tables for depth $P = 7$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	-	-	-	10 [20]
Fixed Angle†	-	-	-	10 [20]
Fourier*	8 [20]	-	8 [20]	10 [20]
Interp.*	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-
Recursive TS	7 [20]	-	8 [20]	9 [20]
TQA	-	-	-	10 [20]
TQA*	-	-	-	10 [20]

Table 49: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	17289.5 $\pm$ 51.9	-	-
Interp. [*]	-	16398.9 $\pm$ 190.0	-	-
TQA	-	-	-	-
TQA [*]	-	-	-	-
MPS (Quimb)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	-	-	-
Interp. [*]	-	-	-	-
TQA	-	-	-	-
TQA [*]	-	-	-	-
PP				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	17176.5 $\pm$ 0.0	-	-
Interp. [*]	-	16528.1 $\pm$ 0.0	-	-
TQA	-	-	-	-
TQA [*]	-	-	-	-
SV				
Fixed Angle [*]	-	-	-	3237.2 $\pm$ 47.0
Fixed Angle [†]	-	-	-	1666.7 $\pm$ 10.3
Fourier [*]	4669.9 $\pm$ 598.0	-	4633.9 $\pm$ 0.0	3459.9 $\pm$ 11.8
Interp. [*]	4003.3 $\pm$ 497.3	-	2811.0 $\pm$ 0.0	3462.4 $\pm$ 2.2
Linear Ramp	-	-	-	-
Recursive TS	4521.0 $\pm$ 596.3	-	4467.6 $\pm$ 0.0	3466.4 $\pm$ 13.2
TQA	-	-	-	3256.5 $\pm$ 3.3
TQA [*]	-	-	-	3492.1 $\pm$ 1.1

Table 50: **Avg. Energy $\pm$ standard deviation for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle [*]	-	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-	-
Fourier [*]	-	17289.5 $\pm$ 51.9	-	-	-	10 [144]	-	-
Interp. [*]	-	16398.9 $\pm$ 190.0	-	-	-	10 [144]	-	-
TQA	-	-	-	-	-	-	-	-
TQA [*]	-	-	-	-	-	-	-	-
MPS (Quimb)								
Fixed Angle [*]	-	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-	-
Fourier [*]	-	-	-	-	-	-	-	-
Interp. [*]	-	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-	-
TQA [*]	-	-	-	-	-	-	-	-
PP								
Fixed Angle [*]	-	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-	-
Fourier [*]	-	17176.5 $\pm$ 0.0	-	-	-	10 [144]	-	-
Interp. [*]	-	16528.1 $\pm$ 0.0	-	-	-	10 [144]	-	-
TQA	-	-	-	-	-	-	-	-
TQA [*]	-	-	-	-	-	-	-	-
SV								
Fixed Angle [*]	-	-	-	3237.2 $\pm$ 47.0	-	-	-	10 [20]
Fixed Angle [†]	-	-	-	1666.7 $\pm$ 10.3	-	-	-	10 [20]
Fourier [*]	4669.9 $\pm$ 598.0	-	4633.9 $\pm$ 0.0	3459.9 $\pm$ 11.8	8 [20]	-	8 [20]	10 [20]
Interp. [*]	4003.3 $\pm$ 497.3	-	2811.0 $\pm$ 0.0	3462.4 $\pm$ 2.2	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-	-	-	-	-
Recursive TS	4521.0 $\pm$ 596.3	-	4467.6 $\pm$ 0.0	3466.4 $\pm$ 13.2	7 [20]	-	8 [20]	9 [20]
TQA	-	-	-	3256.5 $\pm$ 3.3	-	-	-	10 [20]
TQA [*]	-	-	-	3492.1 $\pm$ 1.1	-	-	-	10 [20]

Table 51: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

18 Tables for depth $P = 8$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	-	10 [144]	-	-
Fixed Angle [†]	-	10 [144]	-	-
Fourier [*]	-	10 [144]	-	-
Interp. [*]	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA [*]	-	10 [144]	-	-
MPS (Quimb)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	-	-	-
Interp. [*]	-	-	-	-
TQA	-	10 [144]	-	-
TQA [*]	-	10 [144]	-	-
PP				
Fixed Angle [*]	-	10 [144]	-	-
Fixed Angle [†]	-	10 [144]	-	-
Fourier [*]	-	10 [144]	-	-
Interp. [*]	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA [*]	-	10 [144]	-	-
SV				
Fixed Angle [*]	-	-	-	10 [20]
Fixed Angle [†]	-	-	-	10 [20]
Fourier [*]	8 [20]	-	8 [20]	10 [20]
Interp. [*]	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-
Recursive TS	7 [20]	-	8 [20]	-
TQA	-	-	-	10 [20]
TQA [*]	-	-	-	10 [20]

Table 52: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	-	16943.8 $\pm$ 41.2	-	-
Fixed Angle [†]	-	14081.1 $\pm$ 15.3	-	-
Fourier [*]	-	17315.2 $\pm$ 34.1	-	-
Interp. [*]	-	16279.0 $\pm$ 420.1	-	-
TQA	-	17132.3 $\pm$ 3.1	-	-
TQA [*]	-	17252.6 $\pm$ 20.1	-	-
MPS (Quimb)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	-	-	-
Interp. [*]	-	-	-	-
TQA	-	16609.5 $\pm$ 2.2	-	-
TQA [*]	-	17070.9 $\pm$ 43.3	-	-
PP				
Fixed Angle [*]	-	16897.0 $\pm$ 0.0	-	-
Fixed Angle [†]	-	12075.1 $\pm$ 0.0	-	-
Fourier [*]	-	17161.2 $\pm$ 0.0	-	-
Interp. [*]	-	16582.6 $\pm$ 0.0	-	-
TQA	-	16788.3 $\pm$ 0.0	-	-
TQA [*]	-	17137.5 $\pm$ 0.1	-	-
SV				
Fixed Angle [*]	-	-	-	3253.9 $\pm$ 440.8
Fixed Angle [†]	-	-	-	2042.3 $\pm$ 32.5
Fourier [*]	4678.1 $\pm$ 599.4	-	4600.3 $\pm$ 0.0	3447.5 $\pm$ 18.4
Interp. [*]	3984.0 $\pm$ 507.3	-	2767.6 $\pm$ 0.0	3447.1 $\pm$ 6.4
Linear Ramp	-	-	-	-
Recursive TS	4527.5 $\pm$ 591.3	-	4467.8 $\pm$ 0.0	-
TQA	-	-	-	3331.9 $\pm$ 3.3
TQA [*]	-	-	-	3461.8 $\pm$ 7.9

Table 53: **Avg. Energy $\pm$ standard deviation for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	-	16943.8 $\pm$ 41.2	-	-	-	10 [144]	-	-
Fixed Angle [†]	-	14081.1 $\pm$ 15.3	-	-	-	10 [144]	-	-
Fourier*	-	17315.2 $\pm$ 34.1	-	-	-	10 [144]	-	-
Interp.*	-	16279.0 $\pm$ 420.1	-	-	-	10 [144]	-	-
TQA	-	17132.3 $\pm$ 3.1	-	-	-	10 [144]	-	-
TQA*	-	17252.6 $\pm$ 20.1	-	-	-	10 [144]	-	-
MPS (Quimb)								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-	-
Fourier*	-	-	-	-	-	-	-	-
Interp.*	-	-	-	-	-	-	-	-
TQA	-	16609.5 $\pm$ 2.2	-	-	-	10 [144]	-	-
TQA*	-	17070.9 $\pm$ 43.3	-	-	-	10 [144]	-	-
PP								
Fixed Angle*	-	16897.0 $\pm$ 0.0	-	-	-	10 [144]	-	-
Fixed Angle [†]	-	12075.1 $\pm$ 0.0	-	-	-	10 [144]	-	-
Fourier*	-	17161.2 $\pm$ 0.0	-	-	-	10 [144]	-	-
Interp.*	-	16582.6 $\pm$ 0.0	-	-	-	10 [144]	-	-
TQA	-	16788.3 $\pm$ 0.0	-	-	-	10 [144]	-	-
TQA*	-	17137.5 $\pm$ 0.1	-	-	-	10 [144]	-	-
SV								
Fixed Angle*	-	-	-	3253.9 $\pm$ 440.8	-	-	-	10 [20]
Fixed Angle [†]	-	-	-	2042.3 $\pm$ 32.5	-	-	-	10 [20]
Fourier*	4678.1 $\pm$ 599.4	-	4600.3 $\pm$ 0.0	3447.5 $\pm$ 18.4	8 [20]	-	8 [20]	10 [20]
Interp.*	3984.0 $\pm$ 507.3	-	2767.6 $\pm$ 0.0	3447.1 $\pm$ 6.4	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-	-	-	-	-
Recursive TS	4527.5 $\pm$ 591.3	-	4467.8 $\pm$ 0.0	-	7 [20]	-	8 [20]	-
TQA	-	-	-	3331.9 $\pm$ 3.3	-	-	-	10 [20]
TQA*	-	-	-	3461.8 $\pm$ 7.9	-	-	-	10 [20]

Table 54: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

19 Tables for depth $P = 9$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fixed Angle*	-	-	-	10 [20]
Fixed Angle†	-	-	-	10 [20]
Fourier*	8 [20]	-	8 [20]	10 [20]
Interp.*	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-
Recursive TS	7 [20]	-	8 [20]	-
TQA	-	-	-	10 [20]
TQA*	-	-	-	10 [20]

Table 55: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	17318.8 $\pm$ 27.4	-	-
Interp. [*]	-	16162.6 $\pm$ 379.1	-	-
TQA	-	-	-	-
TQA [*]	-	-	-	-
MPS (Quimb)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	-	-	-
Interp. [*]	-	-	-	-
TQA	-	-	-	-
TQA [*]	-	-	-	-
PP				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	17132.5 $\pm$ 0.0	-	-
Interp. [*]	-	16628.0 $\pm$ 0.0	-	-
TQA	-	-	-	-
TQA [*]	-	-	-	-
SV				
Fixed Angle [*]	-	-	-	3032.3 $\pm$ 655.3
Fixed Angle [†]	-	-	-	1161.7 $\pm$ 22.8
Fourier [*]	4699.4 $\pm$ 624.6	-	4612.7 $\pm$ 0.0	3442.1 $\pm$ 19.5
Interp. [*]	3934.2 $\pm$ 469.5	-	3724.0 $\pm$ 0.0	3441.1 $\pm$ 8.1
Linear Ramp	-	-	-	-
Recursive TS	4527.5 $\pm$ 591.3	-	4467.9 $\pm$ 0.0	-
TQA	-	-	-	3304.2 $\pm$ 4.3
TQA [*]	-	-	-	3465.1 $\pm$ 8.1

Table 56: **Avg. Energy $\pm$ standard deviation for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-	-
Fourier*	-	17318.8 ± 27.4	-	-	-	10 [144]	-	-
Interp.*	-	16162.6 ± 379.1	-	-	-	10 [144]	-	-
TQA	-	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-	-
MPS (Quimb)								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-	-
Fourier*	-	-	-	-	-	-	-	-
Interp.*	-	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-	-
PP								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle†	-	-	-	-	-	-	-	-
Fourier*	-	17132.5 ± 0.0	-	-	-	10 [144]	-	-
Interp.*	-	16628.0 ± 0.0	-	-	-	10 [144]	-	-
TQA	-	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-	-
SV								
Fixed Angle*	-	-	-	3032.3 ± 655.3	-	-	-	10 [20]
Fixed Angle†	-	-	-	1161.7 ± 22.8	-	-	-	10 [20]
Fourier*	4699.4 ± 624.6	-	4612.7 ± 0.0	3442.1 ± 19.5	8 [20]	-	8 [20]	10 [20]
Interp.*	3934.2 ± 469.5	-	3724.0 ± 0.0	3441.1 ± 8.1	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-	-	-	-	-
Recursive TS	4527.5 ± 591.3	-	4467.9 ± 0.0	-	7 [20]	-	8 [20]	-
TQA	-	-	-	3304.2 ± 4.3	-	-	-	10 [20]
TQA*	-	-	-	3465.1 ± 8.1	-	-	-	10 [20]

Table 57: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

20 Tables for depth $P = 10$

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle*	-	10 [144]	-	-
Fixed Angle†	-	10 [144]	-	-
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA*	-	10 [144]	-	-
MPS (Quimb)				
Fixed Angle*	-	-	-	-
Fixed Angle†	-	-	-	-
Fourier*	-	-	-	-
Interp.*	-	-	-	-
TQA	-	10 [144]	-	-
TQA*	-	10 [144]	-	-
PP				
Fixed Angle*	-	10 [144]	-	-
Fixed Angle†	-	10 [144]	-	-
Fourier*	-	10 [144]	-	-
Interp.*	-	10 [144]	-	-
TQA	-	10 [144]	-	-
TQA*	-	10 [144]	-	-
SV				
Fixed Angle*	-	-	-	10 [20]
Fixed Angle†	-	-	-	10 [20]
Fourier*	8 [20]	-	8 [20]	10 [20]
Interp.*	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-
Recursive TS	7 [20]	-	8 [20]	1 [20]
TQA	-	-	-	10 [20]
TQA*	-	-	-	10 [20]

Table 58: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	LB	RR
MPS (Aer)				
Fixed Angle [*]	-	16563.8 $\pm$ 349.2	-	-
Fixed Angle [†]	-	14320.0 $\pm$ 9.5	-	-
Fourier [*]	-	17326.1 $\pm$ 29.2	-	-
Interp. [*]	-	16072.7 $\pm$ 324.4	-	-
TQA	-	17167.3 $\pm$ 2.4	-	-
TQA [*]	-	17298.2 $\pm$ 14.8	-	-
MPS (Quimb)				
Fixed Angle [*]	-	-	-	-
Fixed Angle [†]	-	-	-	-
Fourier [*]	-	-	-	-
Interp. [*]	-	-	-	-
TQA	-	16800.2 $\pm$ 1.5	-	-
TQA [*]	-	17178.2 $\pm$ 11.7	-	-
PP				
Fixed Angle [*]	-	16929.1 $\pm$ 0.0	-	-
Fixed Angle [†]	-	12444.4 $\pm$ 0.0	-	-
Fourier [*]	-	17128.7 $\pm$ 0.0	-	-
Interp. [*]	-	16575.6 $\pm$ 0.0	-	-
TQA	-	16835.4 $\pm$ 0.0	-	-
TQA [*]	-	17142.7 $\pm$ 0.0	-	-
SV				
Fixed Angle [*]	-	-	-	3382.9 $\pm$ 19.6
Fixed Angle [†]	-	-	-	2014.8 $\pm$ 26.6
Fourier [*]	4717.4 $\pm$ 646.6	-	4659.3 $\pm$ 0.0	3414.5 $\pm$ 35.6
Interp. [*]	3972.9 $\pm$ 494.5	-	3995.4 $\pm$ 0.0	3418.3 $\pm$ 11.3
Linear Ramp	-	-	-	-
Recursive TS	4527.5 $\pm$ 591.3	-	4467.9 $\pm$ 0.0	3447.7
TQA	-	-	-	3277.5 $\pm$ 2.6
TQA [*]	-	-	-	3463.6 $\pm$ 9.6

Table 59: **Avg. Energy $\pm$ standard deviation for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better energies. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy				Num. Instances			
	ER	HH	LB	RR	ER	HH	LB	RR
MPS (Aer)								
Fixed Angle*	-	16563.8 $\pm$ 349.2	-	-	-	10 [144]	-	-
Fixed Angle [†]	-	14320.0 $\pm$ 9.5	-	-	-	10 [144]	-	-
Fourier*	-	17326.1 $\pm$ 29.2	-	-	-	10 [144]	-	-
Interp.*	-	16072.7 $\pm$ 324.4	-	-	-	10 [144]	-	-
TQA	-	17167.3 $\pm$ 2.4	-	-	-	10 [144]	-	-
TQA*	-	17298.2 $\pm$ 14.8	-	-	-	10 [144]	-	-
MPS (Quimb)								
Fixed Angle*	-	-	-	-	-	-	-	-
Fixed Angle [†]	-	-	-	-	-	-	-	-
Fourier*	-	-	-	-	-	-	-	-
Interp.*	-	-	-	-	-	-	-	-
TQA	-	16800.2 $\pm$ 1.5	-	-	-	10 [144]	-	-
TQA*	-	17178.2 $\pm$ 11.7	-	-	-	10 [144]	-	-
PP								
Fixed Angle*	-	16929.1 $\pm$ 0.0	-	-	-	10 [144]	-	-
Fixed Angle [†]	-	12444.4 $\pm$ 0.0	-	-	-	10 [144]	-	-
Fourier*	-	17128.7 $\pm$ 0.0	-	-	-	10 [144]	-	-
Interp.*	-	16575.6 $\pm$ 0.0	-	-	-	10 [144]	-	-
TQA	-	16835.4 $\pm$ 0.0	-	-	-	10 [144]	-	-
TQA*	-	17142.7 $\pm$ 0.0	-	-	-	10 [144]	-	-
SV								
Fixed Angle*	-	-	-	3382.9 $\pm$ 19.6	-	-	-	10 [20]
Fixed Angle [†]	-	-	-	2014.8 $\pm$ 26.6	-	-	-	10 [20]
Fourier*	4717.4 $\pm$ 646.6	-	4659.3 $\pm$ 0.0	3414.5 $\pm$ 35.6	8 [20]	-	8 [20]	10 [20]
Interp.*	3972.9 $\pm$ 494.5	-	3995.4 $\pm$ 0.0	3418.3 $\pm$ 11.3	7 [20]	-	8 [20]	10 [20]
Linear Ramp	-	-	-	-	-	-	-	-
Recursive TS	4527.5 $\pm$ 591.3	-	4467.9 $\pm$ 0.0	3447.7	7 [20]	-	8 [20]	1 [20]
TQA	-	-	-	3277.5 $\pm$ 2.6	-	-	-	10 [20]
TQA*	-	-	-	3463.6 $\pm$ 9.6	-	-	-	10 [20]

Table 60: **Avg. Energy and Number of Instances for Various Evaluation Methods for MIS for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (LB), and Random Regular (RR). Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. The minimum and maximum energies defining the range of colors are shared between MPS and PP. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.